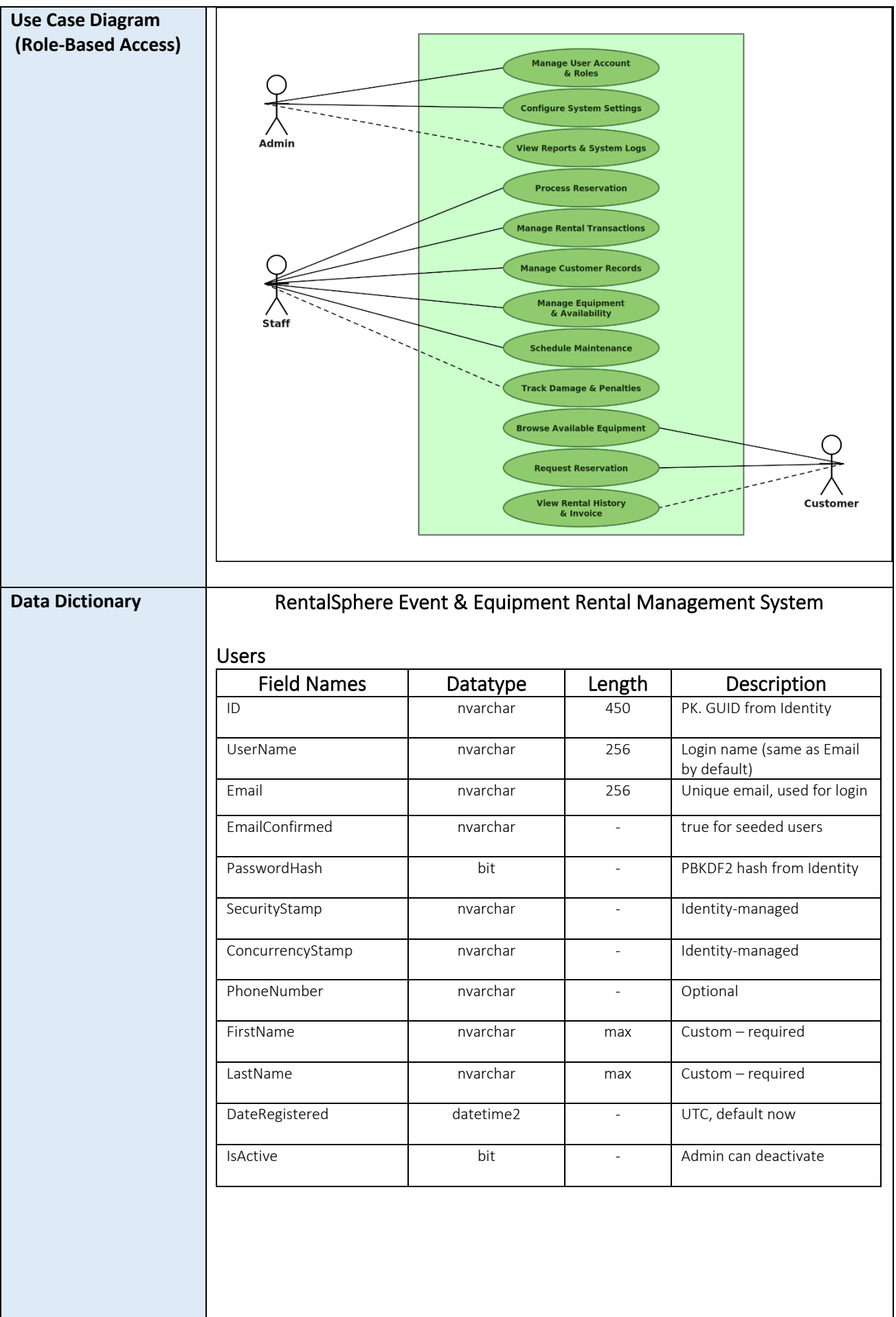


| FINAL PROJECT DOCUMENTATION      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME                             | ANGELO RANES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| PROJECT TITLE                    | <b>RentalSphere: ERP and CRM System for Event &amp; Party Equipment Rental Businesses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SUBJECT:<br>CODE:<br>TIME:       | IT15/L Integrative Programming and Technologies<br>3481<br>8:00AM – 10:00AM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| TOPIC (Type of Business Process) | #30 Rental Management Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Products/Services                | Event & Party Equipment Rental Services                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Website/Deployed (Link)          | <a href="http://rentalsphere.runasp.net">http://rentalsphere.runasp.net</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| API/ALGO/MODEL:                  | <p><b>RentalSphere – APIs / Algorithms / Models Used</b></p> <p><b>Frameworks &amp; tools</b></p> <ol style="list-style-type: none"> <li>1. ASP.NET Core 10 MVC – web application framework.</li> <li>2. Entity Framework Core 10 (Code-First) – ORM.</li> <li>3. Microsoft SQL Server (managed instance in production; LocalDB for development).</li> <li>4. ASP.NET Core Identity – authentication &amp; role-based authorization.</li> <li>5. Bootstrap 5 + Razor Views – server-rendered UI.</li> </ol> <p><b>Authentication &amp; Security algorithms</b></p> <ol style="list-style-type: none"> <li>1. PBKDF2 password hashing (built into Identity).</li> <li>2. Signed, HttpOnly, Secure cookie (Identity.Application) for session.</li> <li>3. AntiForgeryToken validation on every POST (CSRF protection).</li> </ol> <p><b>Architectural patterns &amp; design models</b></p> <ol style="list-style-type: none"> <li>1. Layered architecture: Controller -&gt; Service -&gt; Repository -&gt; DbContext.</li> <li>2. DTO / ViewModel pattern – EF entities never exposed to views.</li> <li>3. Defense-in-depth RBAC: [Authorize(Roles=...)] on controllers, ICurrentUser service guard, query-level customer scoping.</li> <li>4. Append-only audit log – every privileged action writes actor, action, entity, old/new JSON, timestamp, IP.</li> </ol> <p><b>Business algorithms &amp; processes</b></p> <ol style="list-style-type: none"> <li>1. Rental transaction workflow – single record mutates through Active -&gt; Returned -&gt; (optional) Cancelled; return updates the same row (no new record).</li> <li>2. Pricing – DailyRate x days x quantity; configurable late fee per day applied on late return; optional damage penalty added when return is flagged.</li> <li>3. Availability check – date-range intersection query across EquipmentItem + active RentalTransaction to count free units.</li> <li>4. Invoice numbering – INV-YYYYMM-##### (monthly sequential counter).</li> <li>5. Damage-penalty &lt;-&gt; invoice sync – SyncDamageLineAsync keeps the linked InvoiceLine amount in sync with the DamagePenalty record, recomputing invoice totals.</li> <li>6. Cancellation rollback – when a Confirmed reservation is cancelled, its reserved equipment items are released back to Available.</li> </ol> |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>7. <b>Idempotent seeder</b> – default roles &amp; users created only if missing; safe to run on every startup.</p> <p>External integrations – None. All logic runs locally against the SQL Server database.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Security Features</b>                            | <p><b>RentalSphere – Core Security Features</b></p> <ol style="list-style-type: none"> <li>1. <b>Secure Sign-In System</b> – Uses Microsoft’s built-in identity platform; passwords are scrambled, accounts can be turned on/off, and reset links work automatically.</li> <li>2. <b>Three Clear Access Levels</b> – Admin (full control), Staff (day-to-day work), Customer (self-service only). Every screen checks your level before letting you in.</li> <li>3. <b>Admin-Only Dangerous Actions</b> – Only an Admin can delete equipment, cancel a rental, waive a fee, manage other users, or view the audit log.</li> <li>4. <b>Customers See Only Their Own Data</b> – A customer can view only their reservations, invoices, profile, and notes—never anyone else’s.</li> <li>5. <b>Tamper-Proof Activity Log</b> – Every important change (who did what, when, from which IP, old vs. new values) is written once and can’t be altered; only Admins can read it.</li> <li>6. <b>Protected Login Cookie</b> – The session cookie is HTTP-only, SameSite, and sent securely (HTTPS) or safely on HTTP; it expires after 8 hours of inactivity.</li> <li>7. <b>HTTPS Enforced in Production</b> – All live traffic is forced to HTTPS and the browser remembers it (HSTS).</li> <li>8. <b>CSRF Guards on Every Form</b> – Every button that changes data carries a hidden token so a malicious site can’t forge a request.</li> <li>9. <b>No Raw SQL</b> – Injection-Proof – All database calls go through Entity Framework, which automatically parameterises queries.</li> <li>10. <b>Friendly Error Pages, Detailed Server Logs</b> – Users get a clean “403/404” page; developers see the full stack trace only in the server logs.</li> </ol> |
| <b>Target user/s:</b>                               | Admin, Staff, Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Subs System / Management Transaction/Modules</b> | <ol style="list-style-type: none"> <li>1. Equipment Management Module</li> <li>2. Customer Management Module</li> <li>3. Reservation Management Module</li> <li>4. Rental Transactions Module</li> <li>5. Equipment Availability Module</li> <li>6. Maintenance Management Module</li> <li>7. Billing Module</li> <li>8. Customer CRM Module</li> <li>9. Damage &amp; Penalty Tracking Module</li> <li>10. Reports Module</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LOGO</b>                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Project Objectives:</b>              | <ol style="list-style-type: none"> <li>1. To design and develop a centralized web-based ERP and CRM system for event and party equipment rental businesses using ASP.NET Core MVC within the current term.</li> <li>2. To automate reservation and equipment availability tracking, reducing double booking and manual scheduling conflicts.</li> <li>3. To implement a role-based billing and transaction module that accurately records rental fees, damage penalties, and payment status per reservation.</li> <li>4. To provide management with real-time reports on equipment usage, maintenance status, and revenue through a centralized dashboard.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project Description:</b>             | <p>RentalSphere is proposed for small to mid-sized event and party equipment rental businesses that currently rely on manual logs, spreadsheets, or disconnected tools to manage bookings, inventory, and billing. Its target users include rental staff who process reservations and transactions, inventory/maintenance officers who track equipment condition and availability, administrators who oversee operations and reporting, and customers who can browse and reserve equipment online.</p> <p>By centralizing these processes, RentalSphere aims to reduce scheduling conflicts, improve equipment turnaround visibility, and give business owners clearer financial and operational reporting, ultimately supporting more reliable service for customers and less manual workload for staff.</p> <p>The system will be built on ASP.NET Core MVC using C#, with Entity Framework Core handling data access and Microsoft SQL Server serving as the database. The front end will use Razor Views with Bootstrap for a responsive interface, developed in Visual Studio Code.</p> <p>This stack was chosen for its strong tooling support, built-in security features for authentication and role-based access, and scalability for handling multiple concurrent reservations and transactions, aligning with the course's required .NET technology stack.</p> |
| <b>Type of Users/ Role-Based Access</b> | <ol style="list-style-type: none"> <li>1. <b>Admin</b><br/>Username: admin@rentalsphere.local<br/>Password: Admin@12345</li> <li>2. <b>Staff</b><br/>Username: staff@rentalsphere.local<br/>Password: Staff@12345</li> <li>3. <b>Customer</b><br/>Username: customer@rentalsphere.local<br/>Password: Customer@12345</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



## Roles

| Field Names      | Datatype | Length | Description                    |
|------------------|----------|--------|--------------------------------|
| ID               | nvarchar | 450    | PK                             |
| Name             | nvarchar | 256    | One of: Admin, Staff, Customer |
| NormalizedName   | nvarchar | 256    | Identity-managed (uppercase)   |
| ConcurrencyStamp | nvarchar | -      | Identity-managed               |

## UserRoles

| Field Names | Datatype      | Length | Description |
|-------------|---------------|--------|-------------|
| UserID      | Nvarchar (FK) | -      | → Users.Id  |
| RoleID      | Nvarchar (FK) | -      | → Roles.Id  |

## UserClaims

| Field Names | Datatype | Length | Description                                  |
|-------------|----------|--------|----------------------------------------------|
| ID          | int      | -      | PK.Identity                                  |
| UserId      | nvarchar | 450    | FK → Users.Id (Cascade delete)               |
| ClaimType   | nvarchar | -      | Type of claim (e.g., permission, department) |
| ClaimValue  | nvarchar | -      | Value of the claim (e.g., read, sales)       |

## UserLogins

| Field Names         | Datatype | Length | Description                                            |
|---------------------|----------|--------|--------------------------------------------------------|
| LoginProvider       | nvarchar | 128    | PK (composite). Provider name (e.g., Google, Facebook) |
| ProviderKey         | nvarchar | 128    | PK (composite). Provider's unique user ID              |
| ProviderDisplayName | nvarchar | max    | Display name shown on the login UI                     |
| UserId              | nvarchar | 450    | FK → Users.Id (Cascade delete)                         |

## UserTokens

| Field Names   | Datatype | Length | Description                                                                   |
|---------------|----------|--------|-------------------------------------------------------------------------------|
| UserId        | nvarchar | 450    | PK (composite). FK → Users.Id (Cascade delete)                                |
| LoginProvider | nvarchar | 128    | PK (composite). Provider name (e.g., Default, Google)                         |
| Name          | nvarchar | 128    | PK (composite). Token name (e.g., ResetPasswordToken, EmailConfirmationToken) |
| Value         | nvarchar | max    | The actual token string                                                       |

### Categories

| Field Names | Datatype | Length | Description  |
|-------------|----------|--------|--------------|
| CategoryID  | int      | -      | PK. Identity |
| Name        | Nvarchar | max    | Required     |
| Description | Nvarchar | max    | Optional     |

### Equipment

| Field Names   | Datatype      | Length | Description                                 |
|---------------|---------------|--------|---------------------------------------------|
| EquipmentID   | int           | -      | PK. Identity                                |
| Name          | nvarchar      | max    | Required                                    |
| CategoryID    | int           | -      | FK → Categories                             |
| Description   | nvarchar      | max    | Optional                                    |
| DailyRate     | decimal(18,2) | -      | ₱ per day per unit                          |
| StockQuantity | int           | -      | Denormalized count of Available items       |
| Status        | int           | -      | enum: 0=Available, 1=Unavailable, 2=Retired |
| DateAdded     | datetime2     | -      | UTC, default now                            |
| ImageURL      | nvarchar      | max    | Optional photo URL                          |

### EquipmentItems

| Field Names        | Datatype  | Length | Description                                                             |
|--------------------|-----------|--------|-------------------------------------------------------------------------|
| ItemID             | int       | -      | PK. Identity                                                            |
| EquipmentID        | int       | -      | FK → Equipment                                                          |
| SerialNumber       | nvarchar  | max    | Required, unique per EquipmentID                                        |
| AvailabilityStatus | int       | -      | enum: 0=Available, 1=Reserved, 2=CheckedOut, 3=InMaintenance, 4=Retired |
| Location           | nvarchar  | max    | Optional warehouse/bin                                                  |
| LastStatusChange   | datetime2 | -      | Stamped on every transition                                             |

### Customers

| Field Names | Datatype | Length | Description                                  |
|-------------|----------|--------|----------------------------------------------|
| CustomerID  | int      | -      | PK. Identity                                 |
| UserID      | nvarchar | 450    | FK → Users.Id. Nullable for walk-ins         |
| FirstName   | nvarchar | max    | Required                                     |
| LastName    | nvarchar | max    | Required                                     |
| Email       | nvarchar | max    | Required                                     |
| Phone       | nvarchar | max    | Required                                     |
| Address     | nvarchar | max    | Required                                     |
| City        | nvarchar | max    | Required                                     |
| PostalCode  | nvarchar | max    | Required                                     |
| LoyaltyTier | int      | -      | enum: 0=Bronze, 1=Silver, 2=Gold, 3=Platinum |

|  |                           |                 |               |                                                                      |
|--|---------------------------|-----------------|---------------|----------------------------------------------------------------------|
|  | TotalSpent                | decimal(18,2)   | -             | Denormalized, displayed on profile                                   |
|  | DateRegistered            | datetime2       | -             | UTC, default now                                                     |
|  | IsActive                  | bit             | -             | Default true; only Admin may flip                                    |
|  | <b>Reservations</b>       |                 |               |                                                                      |
|  | <b>Field Names</b>        | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                                   |
|  | ReservationID             | int             | -             | PK. Identity                                                         |
|  | CustomerID                | int             | -             | FK → Customers                                                       |
|  | ReservationDate           | datetime2       | -             | UTC, default now                                                     |
|  | RentalStartDate           | datetime2       | -             | Required                                                             |
|  | RentalEndDate             | datetime2       | -             | Required; validated > start                                          |
|  | Status                    | int             | -             | enum: 0=Pending, 1=Confirmed, 2=Cancelled, 3=Expired                 |
|  | TotalEstimatedCost        | decimal(18,2)   | -             | $\Sigma(\text{DailyRate} \times \text{days} \times \text{Quantity})$ |
|  | Notes                     | nvarchar        | max           | Optional                                                             |
|  | CreatedByUserId           | nvarchar        | 450           | FK → Users.Id – staff/admin who created                              |
|  | <b>ReservationItems</b>   |                 |               |                                                                      |
|  | <b>Field Names</b>        | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                                   |
|  | ReservationItemID         | int             | -             | PK. Identity                                                         |
|  | ReservationID             | int             | -             | FK → Reservations (Cascade)                                          |
|  | EquipmentID               | int             | -             | FK → Equipment                                                       |
|  | Quantity                  | int             | -             | Range 1–10000                                                        |
|  | AssignedItemID            | int             | -             | FK → EquipmentItems (SetNull). Set at Confirm                        |
|  | Notes                     | nvarchar        | max           | Optional                                                             |
|  | <b>RentalTransactions</b> |                 |               |                                                                      |
|  | <b>Field Names</b>        | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                                   |
|  | RentalTransactionID       | int             | -             | PK. Identity                                                         |
|  | ReservationID             | int             | -             | FK → Reservations (SetNull). Nullable for walk-ins                   |
|  | CustomerID                | int             | -             | FK → Customers (snapshot at Checkout)                                |
|  | CheckoutDate              | datetime2       | -             | UTC, default now                                                     |
|  | ExpectedReturnDate        | datetime2       | -             | Required                                                             |
|  | ReturnDate                | datetime2       | -             | Null while Active; set on Return                                     |
|  | ConditionNotes            | nvarchar        | 2000          | Max 2000 chars                                                       |
|  | Status                    | int             | -             | enum: 0=Active, 1=Returned, 2=Cancelled                              |
|  | TotalAmount               | decimal(18,2)   | -             | Snapshot of Reservation.TotalEstimated Cost                          |
|  |                           |                 |               |                                                                      |

|  |                               |                 |               |                                                   |
|--|-------------------------------|-----------------|---------------|---------------------------------------------------|
|  | ProcessedByUserId             | nvarchar        | 450           | FK → Users.Id – checked out by                    |
|  | ReturnedToUserId              | nvarchar        | 450           | FK → Users.Id – return processed by               |
|  | FlaggedForDamage              | bit             | -             | Set true at Return when damage needs follow-up    |
|  | <b>RentalTransactionItems</b> |                 |               |                                                   |
|  | <b>Field Names</b>            | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                |
|  | RentalTransactionItemID       | int             | -             | PK. Identity                                      |
|  | RentalTransactionID           | int             | -             | FK → RentalTransactions (Cascade)                 |
|  | EquipmentID                   | int             | -             | FK → Equipment                                    |
|  | Quantity                      | int             | -             | Range 1–10000                                     |
|  | EquipmentItemID               | int             | -             | FK → EquipmentItems (SetNull). Pinned at Checkout |
|  | <b>MaintenanceRecords</b>     |                 |               |                                                   |
|  | <b>Field Names</b>            | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                |
|  | MaintenanceRecordID           | int             | -             | PK. Identity                                      |
|  | EquipmentItemID               | int             | -             | FK → EquipmentItems                               |
|  | StartedAt                     | datetime2       | -             | UTC, default now                                  |
|  | ExpectedEnd                   | datetime2       | -             | Optional target                                   |
|  | EndedAt                       | datetime2       | -             | Null while open                                   |
|  | Reason                        | nvarchar        | 500           | Required                                          |
|  | Notes                         | nvarchar        | 2000          | Optional                                          |
|  | Cost                          | decimal(18,2)   | -             | Maintenance cost                                  |
|  | Status                        | int             | -             | enum: 0=InProgress, 1=Completed                   |
|  | OpenedByUserId                | nvarchar        | 450           | FK → Users.Id                                     |
|  | ClosedByUserId                | nvarchar        | 450           | FK → Users.Id                                     |
|  | <b>Invoices</b>               |                 |               |                                                   |
|  | <b>Field Names</b>            | <b>Datatype</b> | <b>Length</b> | <b>Description</b>                                |
|  | InvoiceID                     | int             | -             | PK. Identity                                      |
|  | RentalTransactionID           | int             | -             | FK → RentalTransactions (Restrict)                |
|  | CustomerID                    | int             | -             | FK → Customers (Restrict, denormalized)           |
|  | InvoiceNumber                 | nvarchar        | 20            | Format INV-YYYYMM-####, unique                    |
|  | InvoiceDate                   | datetime2       | -             | UTC, default now                                  |
|  | DueDate                       | datetime2       | -             | Default InvoiceDate + 14 days                     |
|  | Status                        | int             | -             | enum: 0=Unpaid, 1=PartiallyPaid, 2=Paid, 3=Voided |
|  | SubTotal                      | decimal(18,2)   | -             | Σ(LineTotal) before waivers                       |

|  |                     |               |        |                                                                |
|--|---------------------|---------------|--------|----------------------------------------------------------------|
|  | TotalAmount         | decimal(18,2) | -      | SubTotal – $\Sigma$ (WaivedAmount)                             |
|  | AmountPaid          | decimal(18,2) | -      | $\Sigma$ (Payment.Amount)                                      |
|  | Notes               | nvarchar      | 1000   | Optional                                                       |
|  | VoidedAt            | datetime2     | -      | Stamped when Status → Voided                                   |
|  | VoidedByUserId      | nvarchar      | 450    | FK → Users.Id                                                  |
|  | VoidedReason        | nvarchar      | 500    | Required to void                                               |
|  | CreatedByUserId     | nvarchar      | 450    | FK → Users.Id                                                  |
|  | <b>InvoiceLines</b> |               |        |                                                                |
|  | Field Names         | Datatype      | Length | Description                                                    |
|  | InvoiceLineID       | int           | -      | PK. Identity                                                   |
|  | InvoiceID           | int           | -      | FK → Invoices (Cascade)                                        |
|  | LineType            | int           | -      | enum: 0=Rental, 1=LateFee, 2=DamagePenalty, 3=ManualAdjustment |
|  | Description         | nvarchar      | -      | Required                                                       |
|  | Quantity            | decimal(10,2) | -      | Default 1                                                      |
|  | UnitAmount          | decimal(10,2) | -      | ₱ per unit                                                     |
|  | LineTotal           | decimal(10,2) | -      | Quantity × UnitAmount (persisted)                              |
|  | WaivedAmount        | decimal(18,2) | -      | Default 0; Admin may waive                                     |
|  | ReferenceId         | int           | -      | Loose FK to source row (RTItem or DamagePenalty)               |
|  | CreatedAt           | datetime2     | -      | UTC, default now                                               |
|  | <b>Payments</b>     |               |        |                                                                |
|  | Field Names         | Datatype      | Length | Description                                                    |
|  | PaymentID           | int           | -      | PK. Identity                                                   |
|  | InvoiceID           | int           | -      | FK → Invoices (Restrict)                                       |
|  | PaymentDate         | datetime2     | -      | UTC, default now                                               |
|  | Amount              | decimal(18,2) | -      | > 0; capped at remaining balance                               |
|  | Method              | int           | -      | enum: 0=Cash, 1=BankTransfer, 2=Check, 3=Other                 |
|  | ReferenceNumber     | nvarchar      | 100    | Check #, transfer ref                                          |
|  | Notes               | nvarchar      | 500    | Optional                                                       |
|  | RecordedByUserId    | nvarchar      | 450    | FK → Users.Id                                                  |

### DamagePenalties

| Field Names         | Datatype      | Length | Description                                                |
|---------------------|---------------|--------|------------------------------------------------------------|
| DamagePenaltyID     | int           | -      | PK. Identity                                               |
| RentalTransactionID | int           | -      | FK → RentalTransactions (Restrict)                         |
| EquipmentID         | int           | -      | FK → Equipment (Restrict). Nullable for tx-level penalties |
| EquipmentItemID     | int           | -      | FK → EquipmentItems (SetNull). Specific unit               |
| PenaltyType         | int           | -      | enum: 0=Damage, 1=LateFee, 2=Loss, 3=Cleaning, 4=Other     |
| Description         | nvarchar      | 500    | Required                                                   |
| Amount              | decimal(10,2) | -      | ≥ 0                                                        |
| WaivedAmount        | decimal(18,2) | -      | Default 0                                                  |
| Status              | int           | -      | enum: 0=Pending, 1=AppliedToInvoice, 2=Waived              |
| ReportedAt          | datetime2     | -      | UTC, default now                                           |
| ReportedByUserId    | nvarchar      | 450    | FK → Users.Id                                              |
| ResolvedAt          | datetime2     | -      | Set when Status → Waived                                   |
| ResolvedByUserId    | nvarchar      | 450    | FK → Users.Id                                              |
| Notes               | nvarchar      | 2000   | Optional                                                   |

### CustomerNotes

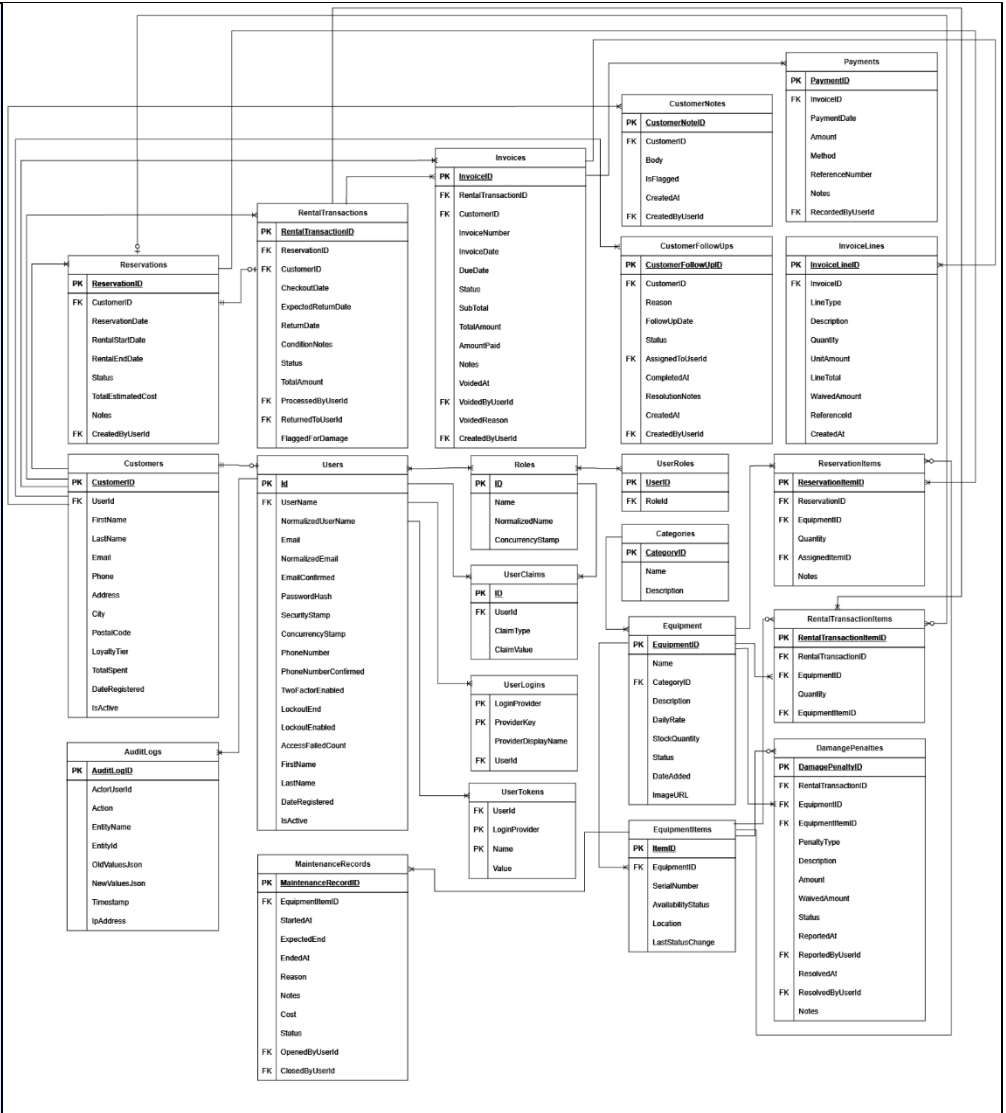
| Field Names     | Datatype  | Length | Description            |
|-----------------|-----------|--------|------------------------|
| CustomerNoteID  | int       | -      | PK. Identity           |
| CustomerID      | int       | -      | FK → Customers         |
| Body            | nvarchar  | max    | Note text              |
| IsFlagged       | bit       | -      | "Needs attention" flag |
| CreatedAt       | datetime2 | -      | UTC, default now       |
| CreatedByUserId | nvarchar  | 450    | FK → Users.Id          |

### CustomerFollowUps

| Field Names        | Datatype  | Length | Description                          |
|--------------------|-----------|--------|--------------------------------------|
| CustomerFollowUpID | int       | -      | PK. Identity                         |
| CustomerID         | int       | -      | FK → Customers                       |
| Reason             | nvarchar  | max    | Required                             |
| FollowUpDate       | datetime2 | -      | UTC – "due" date                     |
| Status             | int       | -      | enum: 0=Pending, 1=Done, 2=Cancelled |
| AssignedToUserId   | nvarchar  | 450    | FK → Users.Id. Null = unassigned     |
| CompletedAt        | datetime2 | -      | Stamped on Done                      |
| ResolutionNotes    | nvarchar  | max    | Optional                             |
| CreatedAt          | datetime2 | -      | UTC, default now                     |
| CreatedByUserId    | nvarchar  | 450    | FK → Users.Id                        |

|  |                    |           |        |                                      |
|--|--------------------|-----------|--------|--------------------------------------|
|  | AuditLogs          |           |        |                                      |
|  | Field Names        | Datatype  | Length | Description                          |
|  | AuditLogID         | int       | -      | PK. Identity                         |
|  | ActorUserId        | nvarchar  | 450    | → Users.Id (no nav property)         |
|  | Action             | nvarchar  | max    | E.g. Customer.Create, Invoice.Void   |
|  | EntityName         | nvarchar  | max    | E.g. Customer, Invoice, InvoiceLine  |
|  | EntityId           | nvarchar  | max    | Stringified PK                       |
|  | OldValuesJson      | nvarchar  | max    | JSON snapshot before                 |
|  | NewValuesJson      | nvarchar  | max    | JSON snapshot after                  |
|  | Timestamp          | datetime2 | -      | UTC, default now                     |
|  | IpAddress          | nvarchar  | max    | Optional                             |
|  | CustomerFollowUps  |           |        |                                      |
|  | Field Names        | Datatype  | Length | Description                          |
|  | CustomerFollowUpID | int       | -      | PK. Identity                         |
|  | CustomerId         | int       | -      | FK → Customers                       |
|  | Reason             | nvarchar  | max    | Required                             |
|  | FollowUpDate       | datetime2 | -      | UTC – "due" date                     |
|  | Status             | int       | -      | enum: 0=Pending, 1=Done, 2=Cancelled |
|  | AssignedToUserId   | nvarchar  | 450    | FK → Users.Id. Null = unassigned     |
|  | CompletedAt        | datetime2 | -      | Stamped on Done                      |
|  | ResolutionNotes    | nvarchar  | max    | Optional                             |
|  | CreatedAt          | datetime2 | -      | UTC, default now                     |
|  | CreatedByUserId    | nvarchar  | 450    | FK → Users.Id                        |

### Entity Relational Diagram (ERD)



Prototype  
(Frontend)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Sign-in screen</b></p> <p>What the user does: Enters their email and password, optionally checks “Remember me”, then clicks Log in. The site checks the credentials, creates a secure login cookie, and redirects the user to the dashboard that matches their role (Admin/Staff get the full menu; Customer sees only their own area). If the credentials are wrong, the same page reloads with an “Invalid login attempt” message.</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

 **Sign in**

Welcome back to RentalSphere.

Email

Password

☐ Remember me

**Sign in**

---

No account? [Register](#)



# Sign in

Welcome back to RentalSphere.

Email

Password

☐ Remember me

[Sign in](#)

---

No account? [Register](#)

## Register screen

What the user does: Clicks Register, fills in email, password and password confirmation, then clicks Create account. The system validates the email isn't already used and that the password meets the minimum rules (8 + characters, upper-/lower-case, a number). On success the account is created, the user is automatically signed in, and taken to the appropriate dashboard. On failure the page shows a clear error (e.g., "Email already registered" or "Passwords do not match").

+ 

## Create an account

Self-registration creates a Customer account. Admins can promote you to Staff or another role.

First name

Last name

Email

Password

Confirm password

Create account

---

Already registered? [Sign in](#)

### Admin Dashboard screen

What the admin sees: After signing in, the admin lands on a summary page that shows key-at-a-glance numbers – total equipment in the catalog, pending reservations, overdue rentals, unpaid invoices, open maintenance jobs, and unread CRM notes. Each number is a clickable card that jumps straight to the corresponding management list (Equipment, Reservations, Transactions, Invoices, Maintenance, CRM). The top navigation bar gives instant access to every module (Equipment, Customers, Reservations, Transactions, Maintenance, Billing, Damage & Penalty, Reports, Settings). Only the Admin role can see the Settings / Audit Log / User-management links.

- RentalSphere**
- Dashboard
- Equipment
- Administration
- People
- Operations
- CRM
- Reports

Type to search...

System Administrator (Admin)

Monday, September 28, 2026

## Dashboard

At-a-glance overview of revenue, rentals, invoices, and follow ups.

**TODAY'S REVENUE**

₹ 0.00

No data yet

[View Invoices →](#)

**ACTIVE RENTALS**

0

Due back this week

[View transactions →](#)

**OPEN INVOICES**

0

No data yet

[Manage billing →](#)

**OVERDUE ITEMS**

0

All caught up!

[Open queue →](#)

**Rentals this week**

Mon-Sun (by checkout date)

| Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|-----|-----|-----|-----|-----|-----|-----|
|     |     |     |     |     |     |     |
|     |     |     |     |     |     |     |
|     |     |     |     |     |     |     |
|     |     |     |     |     |     |     |
|     |     |     |     |     |     |     |

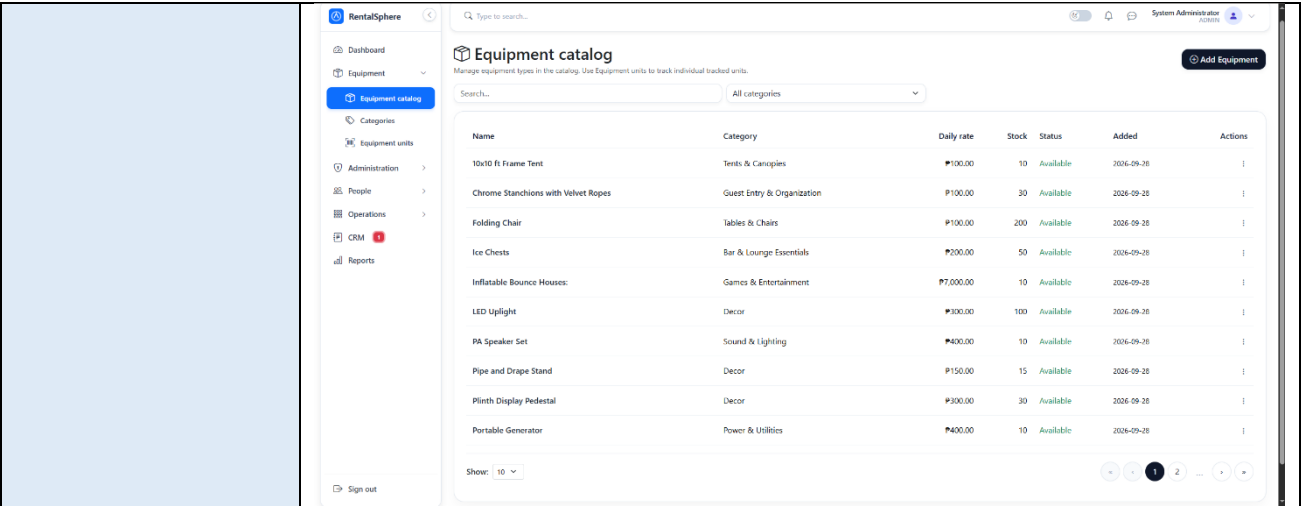
**Recent activity**

- New reservation for Gleen Oliva 5h ago
- New reservation for Gleen Oliva 5h ago

[Sign out](#)
[New Reservation](#)
[Record Payment](#)
[Add Equipment](#)
[View Overdue](#)

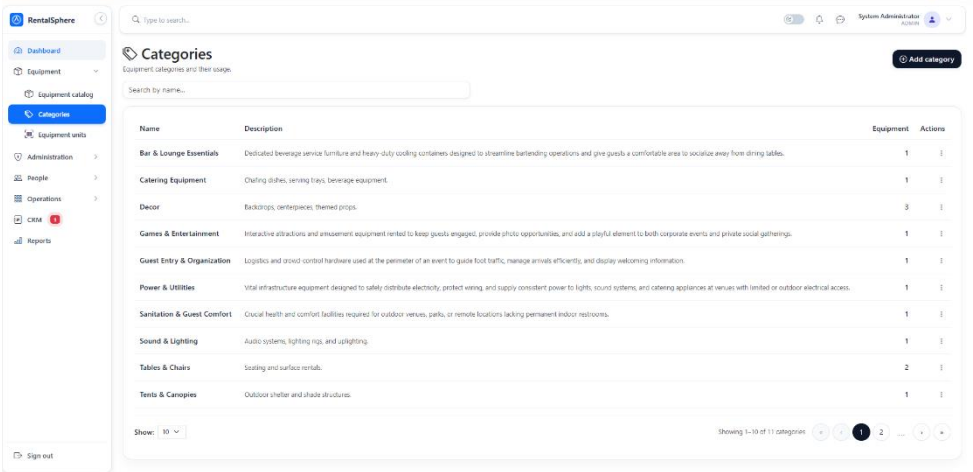
## Equipment Catalog Module

Staff and Admin see a searchable, sortable table with stock-derived status badges and actions to Create, Edit, View, and (Admin only) Delete equipment types; customers see a read-only card grid showing the same badges and a “View details” button. The Create/Edit forms capture name, category, daily rate, total stock (which auto-generates individual item rows), optional description, serial-prefix, and an image. The Details page lists the catalog entry plus every physical Equipment Item with its serial number and live status (Available / Reserved / Checked-Out / Maintenance). Separate Equipment-Item views let Staff change a unit’s status or send it to Maintenance, while only Admin can remove items.



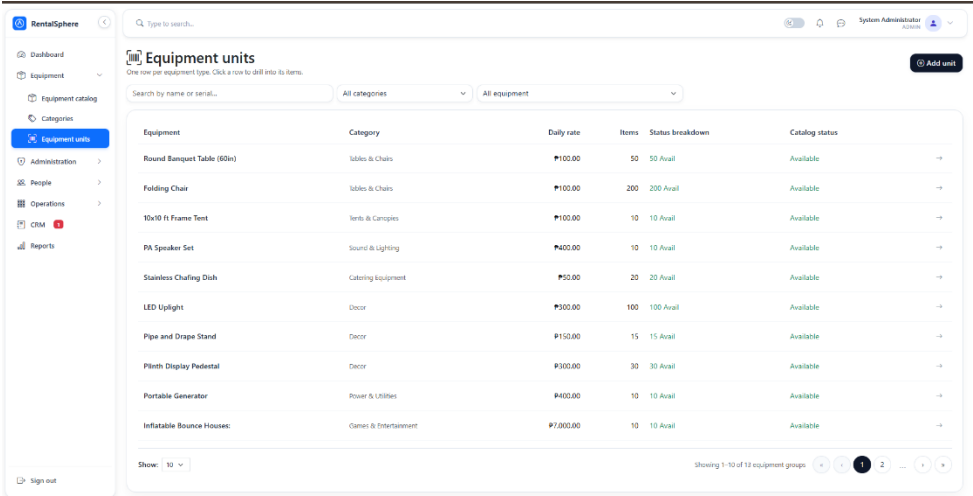
### Categories Module

Staff and Admin see a searchable table of equipment categories (name, description, equipment count) with Create, Edit, and Delete (Admin only) actions. Customers never see this screen. The Create/Edit form only requires a unique name and an optional description; deleting a category is blocked if any equipment still belongs to it.



### Equipment Units (Equipment Items) Module

Staff and Admin view a table of every physical unit showing its serial number, parent equipment, and live status (Available / Reserved / Checked-Out / Maintenance). They can Create new units (auto-generated serial from the catalog prefix), Edit a unit's status or location, and Open a maintenance record for a unit; only Admin may Delete a unit. Customers never see this screen.



### Accounts & Roles Module

Admin-only area that lists every application user with their e-mail, assigned roles, and lockout status. From the list the Admin can Create a new account (sets e-mail, temporary password, and initial role), Edit an existing user's roles (toggle Admin / Staff / Customer) or reset their password, and Delete a

user (soft-delete, preserves audit history). The screen also shows the current role-membership matrix so the Admin can verify that each person has exactly the permissions required by the RBAC rules.

| Email                           | Name                 | Roles    | Status | Registered | Actions |
|---------------------------------|----------------------|----------|--------|------------|---------|
| A.ranes.534279@umindanao.edu.ph | Angelo Ranes         | Customer | Active | 2020-09-28 | [Icons] |
| admin@rentalsphere.local        | System Administrator | Admin    | Active | 2020-09-28 | [Icons] |
| Annie123@gmail.com              | Annie Leonheart      | Customer | Active | 2020-09-28 | [Icons] |
| customer@rentalsphere.local     | Default Customer     | Customer | Active | 2020-09-28 | [Icons] |
| G.leen123@umindanao.edu.ph      | Gleen Olive          | Customer | Active | 2020-09-28 | [Icons] |
| Jason123@gmail.com              | Jason Tatum          | Customer | Active | 2020-09-28 | [Icons] |
| Josh123@gmail.com               | Josh Timber          | Customer | Active | 2020-09-28 | [Icons] |
| Maria12@gmail.com               | Maria Gonzales       | Customer | Active | 2020-09-28 | [Icons] |
| staff@rentalsphere.local        | Staff User           | Staff    | Active | 2020-09-28 | [Icons] |
| Sunkent123@gmail.com            | Sunkent Dose         | Customer | Active | 2020-09-28 | [Icons] |

**Audit Log Module**

Admin-only page that lists every system event in chronological order (who did what, when, and on which record). Each row shows the actor's e-mail/role, the action name (e.g., Reservation.Create, Invoice.Generate, Equipment.Delete), the entity type & ID, and a JSON diff of old vs. new values. The list is searchable by actor, action, entity, or date range and paginated for performance. No edit or delete actions exist – the log is immutable and serves as the definitive compliance trail.

| Timestamp | Actor                      | Action             | Entity      | Entity ID | IP              |
|-----------|----------------------------|--------------------|-------------|-----------|-----------------|
| 931       | G.leen123@umindanao.edu.ph | Reservation Create | Reservation | 1         | 202.107.102.200 |
| 934       | G.leen123@umindanao.edu.ph | Reservation Create | Reservation | 2         | 202.107.102.200 |
| 933       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 10        | 202.107.102.200 |
| 932       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 11        | 202.107.102.200 |
| 931       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 12        | 202.107.102.200 |
| 930       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 13        | 202.107.102.200 |
| 929       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 14        | 202.107.102.200 |
| 928       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 15        | 202.107.102.200 |
| 927       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 16        | 202.107.102.200 |
| 926       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 17        | 202.107.102.200 |
| 925       | admin@rentalsphere.local   | Category Create    | Category    | 10        | 202.107.102.200 |
| 924       | admin@rentalsphere.local   | Category Create    | Category    | 10        | 202.107.102.200 |
| 923       | admin@rentalsphere.local   | Category Create    | Category    | 1         | 202.107.102.200 |
| 922       | admin@rentalsphere.local   | Category Create    | Category    | 2         | 202.107.102.200 |
| 921       | admin@rentalsphere.local   | Category Create    | Category    | 3         | 202.107.102.200 |
| 920       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 1         | 202.107.102.200 |
| 919       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 2         | 202.107.102.200 |
| 918       | admin@rentalsphere.local   | Category Create    | Category    | 1         | 202.107.102.200 |
| 917       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 3         | 202.107.102.200 |
| 916       | admin@rentalsphere.local   | Equipment Create   | Equipment   | 4         | 202.107.102.200 |

**Customer Module**

Staff and Admin see a searchable, sortable table of all customers (name, e-mail, phone, loyalty tier, total spent) with Create, Edit, View, and Delete (Admin only) actions. Customers themselves have a read-only My Profile page showing their details and loyalty status, plus a link to My Reservations; they

cannot see other customers or any management actions.

RentalSphere

Dashboard

Equipment

Administration

People

Customers

Reservations

Operations

CRM

Reports

Type to search...

System Administrator

Customers

Manage customer accounts and loyalty tiers

Search by name or email...

All tiers

| Name             | Email                           | Phone            | Loyalty | Registered | Status | Actions |
|------------------|---------------------------------|------------------|---------|------------|--------|---------|
| Ana Cruz         | ana.cruz@example.test           | +63 919 555 0303 | Bronze  | 2024-11-28 | Active |         |
| Default Customer | customer@rentalsphere.local     | +63 920 555 0404 | Bronze  | 2026-09-28 | Active |         |
| Sunken Dose      | Sunken123@gmail.com             | —                | Bronze  | 2026-09-28 | Active |         |
| Maria Gonzales   | Maria123@gmail.com              | —                | Bronze  | 2026-09-28 | Active |         |
| Annie Leonheart  | Annie123@gmail.com              | —                | Bronze  | 2026-09-28 | Active |         |
| Gleen Oliva      | Gleen123Oliva@umindanao.edu.ph  | —                | Bronze  | 2026-09-28 | Active |         |
| Angelo Rames     | A.rames.534229@umindanao.edu.ph | —                | Bronze  | 2026-09-28 | Active |         |
| Jose Reyes       | jose.reyes@example.test         | +63 918 555 0202 | Bronze  | 2026-03-28 | Active |         |
| Maria Santos     | maria.santos@example.test       | +63 917 555 0101 | Bronze  | 2025-07-28 | Active |         |
| Jason Tatum      | Jason123@gmail.com              | —                | Bronze  | 2026-09-28 | Active |         |
| Josh Timber      | Josh123@gmail.com               | —                | Bronze  | 2026-09-28 | Active |         |

Add customer

**Reservation Module**  
Staff/Admin view a searchable, sortable, paginated table of all reservations with status badges and an “Open” action; they can Create new reservations (choose customer, dates, equipment lines), Confirm pending ones (which locks concrete equipment items), Cancel any non-terminal reservation, and Checkout confirmed reservations to start a rental. Customers see only their own reservations in a card grid (My Reservations) with status, dates, estimated cost and a “View details” button; they can Create new reservations for themselves and Cancel only while the reservation is still Pending. The Details screen shows the full header, line items with assigned serial numbers, and action buttons that respect the RBAC rules (Confirm/Cancel/Checkout/Return).

RentalSphere

Dashboard

Equipment

Administration

People

Customers

Reservations

Operations

CRM

Reports

Type to search...

System Administrator

Reservations

All customer reservations

Search by customer name...

All statuses

| #  | Customer    | Rental dates            | Items | Estimated | Status  | Actions |
|----|-------------|-------------------------|-------|-----------|---------|---------|
| #2 | Gleen Oliva | 2026-09-28 – 2026-09-30 | 1     | ₱200.00   | Expired |         |
| #1 | Gleen Oliva | 2026-09-28 – 2026-09-30 | 1     | ₱600.00   | Expired |         |

New reservation

Sign out

**Availability Module**  
A read-only calendar that lets anyone (customers and staff) pick an equipment type and a date range; the page instantly shows a day-by-day grid with Free / Low / Fully booked counts derived from the same half-open overlap algorithm used when creating or confirming reservations. Customers use it to verify dates before making a reservation; staff can glance at utilisation across the catalog. No edit actions – the view is purely informational.

**RentalSphere**

- Dashboard
- Equipment
- Administration
- People
- Operations
- Availability**
  - Rental Transactions
  - Maintenance
  - Billing
  - Damage & Penalty
- Reports

Type to search...

## Availability calendar

Start
End
Category

09/28/2026
10/12/2026
All categories
Apply

| Equipment                                                                                          | Mon<br>09-28 | Tue<br>09-29 | Wed<br>09-30 | Thu<br>10-01 | Fri<br>10-02 | Sat<br>10-03 | Sun<br>10-04 | Mon<br>10-05 | Tue<br>10-06 | Wed<br>10-07 | Thu<br>10-08 | Fri<br>10-09 | Sat<br>10-10 | Sun<br>10-11 |
|----------------------------------------------------------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| 10x10 R Frame Tent<br><small>Helm &amp; Canopies - 10 in stock</small>                             | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           |
| Chrome Stanchions with Velvet Ropes<br><small>Guard Entry &amp; Organization - 30 in stock</small> | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           |
| Folding Chair<br><small>Tables &amp; Chairs - 200 in stock</small>                                 | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          | 200          |
| Ice Chests<br><small>Bar &amp; Lounge Essentials - 50 in stock</small>                             | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           |
| Inflatable Bounce Houses<br><small>Games &amp; Entertainment - 10 in stock</small>                 | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           |
| LED Uplight<br><small>Decor - 100 in stock</small>                                                 | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          |
| PA Speaker Set<br><small>Sound &amp; Lighting - 10 in stock</small>                                | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           |
| Pipe and Drape Stand<br><small>Decor - 15 in stock</small>                                         | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           | 15           |
| Pivott Display Pedestal<br><small>Decor - 30 in stock</small>                                      | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           | 30           |
| Portable Generator<br><small>Power &amp; Utilities - 10 in stock</small>                           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           | 10           |
| Round Banquet Table (60in)<br><small>Tables &amp; Chairs - 50 in stock</small>                     | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           | 50           |
| Soap Dispenser<br><small>Sanitation &amp; Guest Comfort - 100 in stock</small>                     | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          | 100          |
| Stainless Chafing Dish<br><small>Catering Equipment - 20 in stock</small>                          | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           | 20           |

Available (out of stock)
Low stock (1-5 available)
Softest stock (1-5 available)

Counts start from each equipment's stock, then subtract units in maintenance and confirmed/checked-out reservations overlapping each day.

The screenshot displays the RentalSphere web application. On the left is a vertical sidebar menu with options: Dashboard, Equipment, Administration, People, Operations, Availability, Rental Transactions (highlighted), Maintenance, Billing, Damage & Penalty, CRM, Reports, and Sign out. The main content area has a top header bar containing a search input, a user profile dropdown for 'System Administrator', and system status icons. Below the header, the 'Rental Transactions' section is active, featuring a subtitle 'Track checkouts, returns, and active rentals.' and a filter dropdown set to 'All statuses'. A table with columns '#', 'Customer', 'Reservation', 'Checked out', 'Expected return', 'Returned', 'Amount', 'Status', and 'Actions' is present. The table body shows a message: 'No transactions match your filters.' accompanied by a cube icon.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maintenance Module</b><br>Staff and Admin view a table of all maintenance records (open, in-progress, completed) with the equipment serial number, reason, start date, assigned technician, and cost. From any Available or Checked-Out unit they can Open a maintenance record, which immediately flips that unit's status to Maintenance; staff then update notes, add cost, and Close the record, returning the unit to Available. Only Admin can delete a record. Customers never see this module. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The screenshot displays the Rentasphere application interface. On the left is a vertical sidebar menu with options: Dashboard, Equipment, Administration, People, Operations, Availability, Rental transactions, **Maintenance** (highlighted), Billing, Damage & Penalty, CRM, and Reports. At the bottom of the sidebar is a 'Sign out' button.

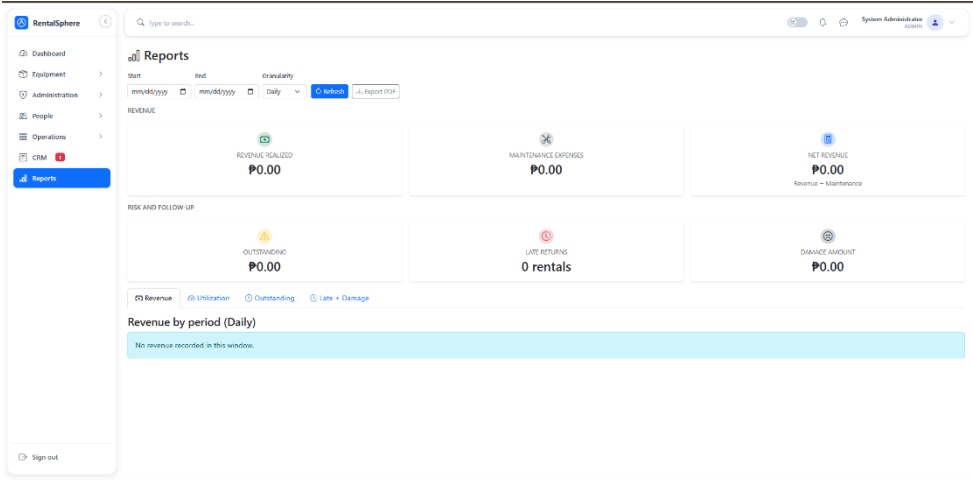
The main header area includes the 'Rentasphere' logo, a search bar with the placeholder 'Type to search...', user information ('System Administrator ASBEN'), and utility icons for settings, notifications, and help. A dark blue button labeled 'Open maintenance.' is located in the top right corner.

The central section is titled 'Maintenance' with a subtitle 'Open tickets (such as repair) and closed records. Staff and Admin only.' Below this is a dropdown menu currently set to 'All statuses'. The main content area features a table with columns: #, Equipment, Serial, Started, Expected, Closed, Reason, Cost, Status, and Actions. The table body contains a large scissor icon and the message 'No maintenance records match your filters.'

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Billing Module</b><br>Staff and Admin see a list of all invoices with status badges (Open, Partial, Paid, Voided) and totals. From a Completed rental the system auto-creates an invoice that includes rental line totals, any late-fee days, and un-waived damage penalties. Staff can Record payments (cash, card, transfer) against an open invoice, view the full payment history, and Void an invoice (Admin only) with a reason. Customers only see their own invoices in a read-only view showing the balance due and a “Print PDF” button. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

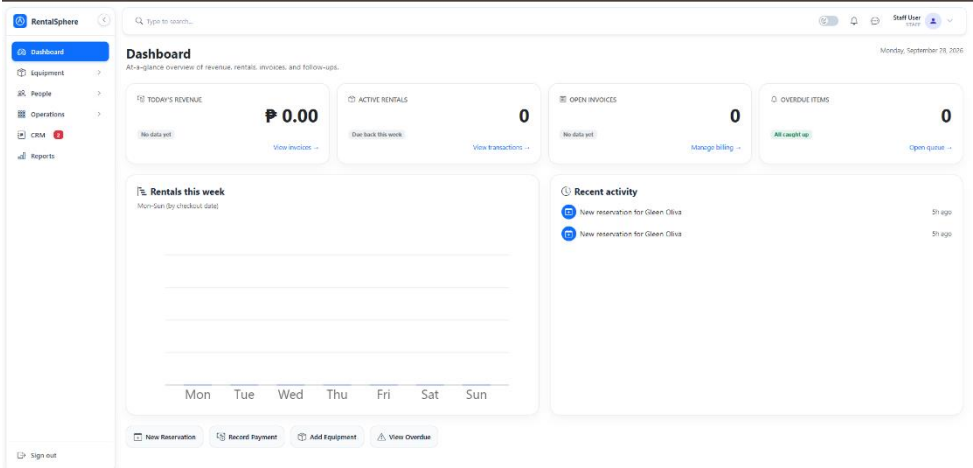


the due date), and Damage Trends (penalty count and amount by equipment and type). Each tab shows a summary card row, a paginated table, and a “Export PDF” button (QuestPDF) for printing or filing. No edit actions – the module is purely analytical.



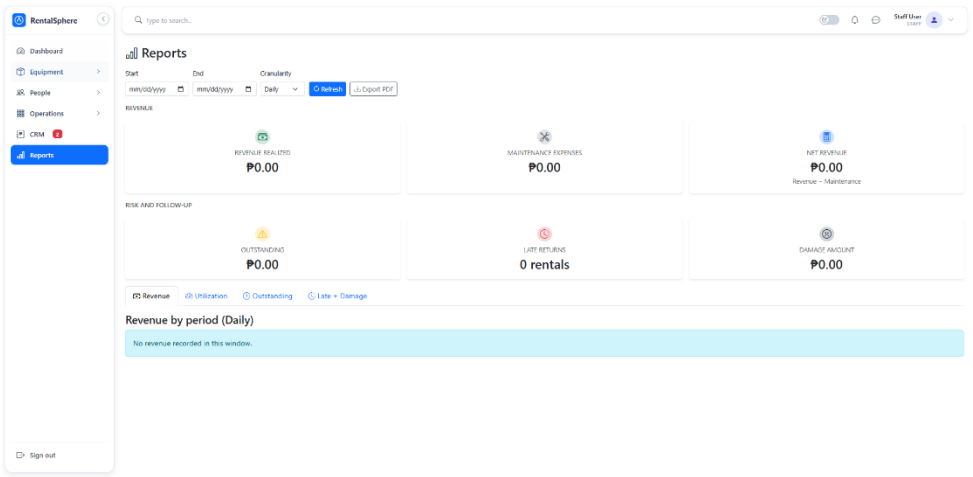
### Staff Dashboard

After signing in, staff land on a summary page that shows the day-to-day workload at a glance: counts of pending reservations, overdue rentals, open maintenance jobs, unpaid invoices, and unread CRM notes. Each count is a clickable card that jumps straight to the corresponding list (Reservations, Rental Transactions, Maintenance, Invoices, CRM). The top navigation bar gives access to every operational module (Equipment, Customers, Reservations, Transactions, Maintenance, Billing, Damage & Penalty, Reports) but hides the Admin-only links (Settings, Audit Log, User-management).



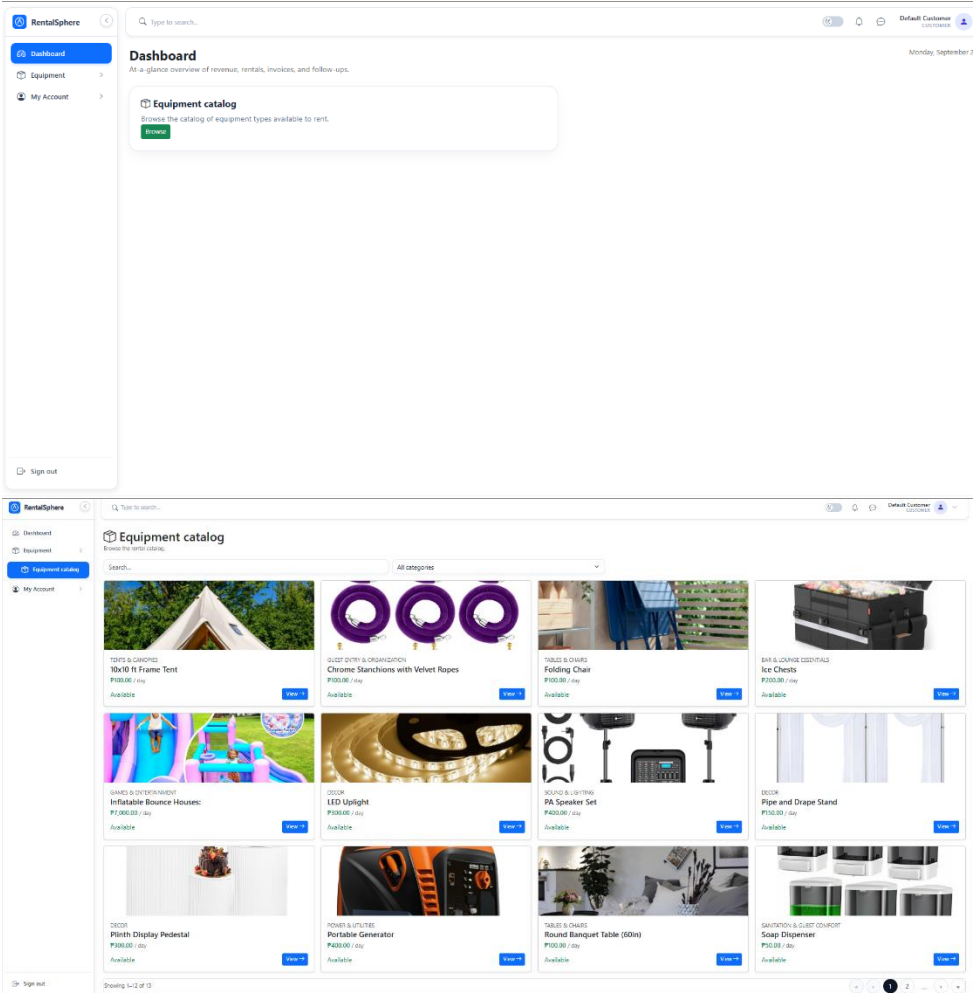
### Staff Reports

Staff can open the Reports area and run the same on-demand tabs as Admin (Revenue, Outstanding, Utilisation, Late Returns, Damage Trends). Each tab shows a summary card row, a paginated table, and an Export PDF button. The only difference is that Staff cannot access the Admin-only settings or audit-log screens; the report data and export capability are identical.



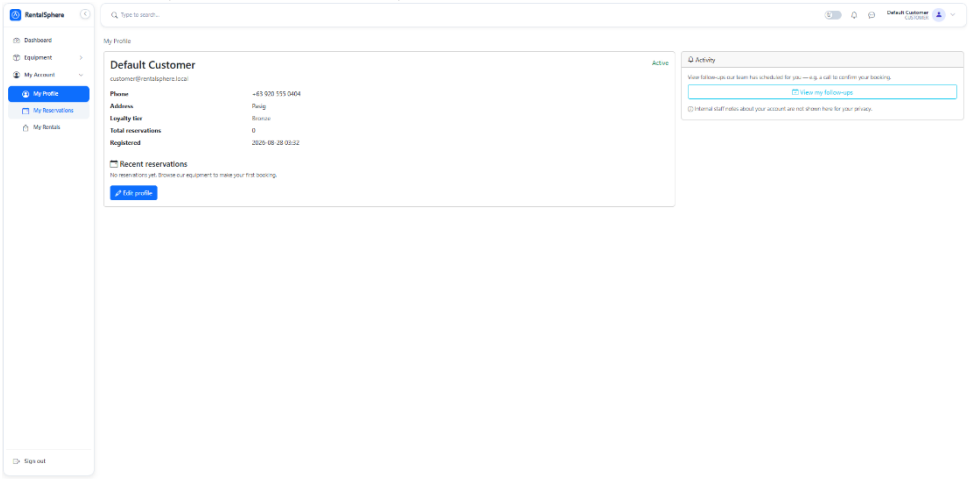
Customer Dashboard → Browse Catalog

The dashboard that appears right after the customer signs in shows a single “Browse Catalog” button. Clicking that button takes the customer to a separate page that displays the full equipment catalog as a responsive card grid—each card shows the item’s photo, category, daily rate, and a live-stock badge (Available / Low Stock / Out of Stock). From there the customer can click View details on any card to see full specs and an availability calendar, and then create a reservation for themselves. No management actions (edit, delete, confirm, etc.) appear anywhere.



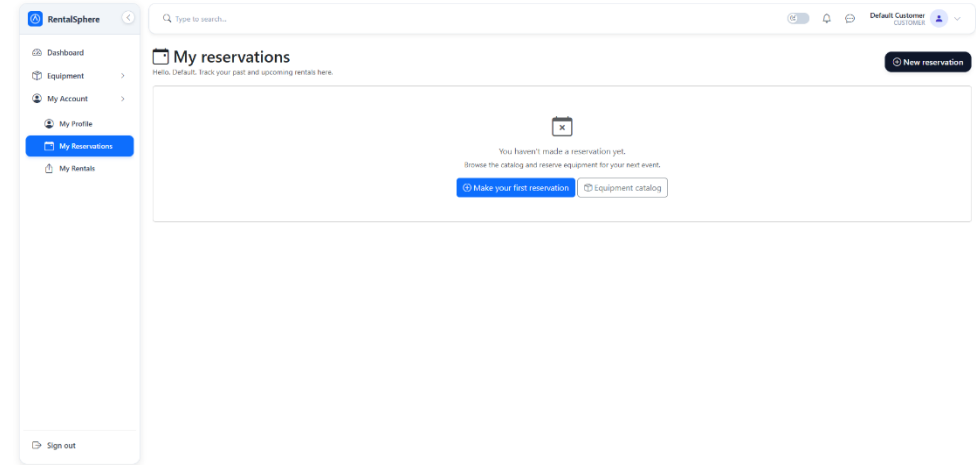
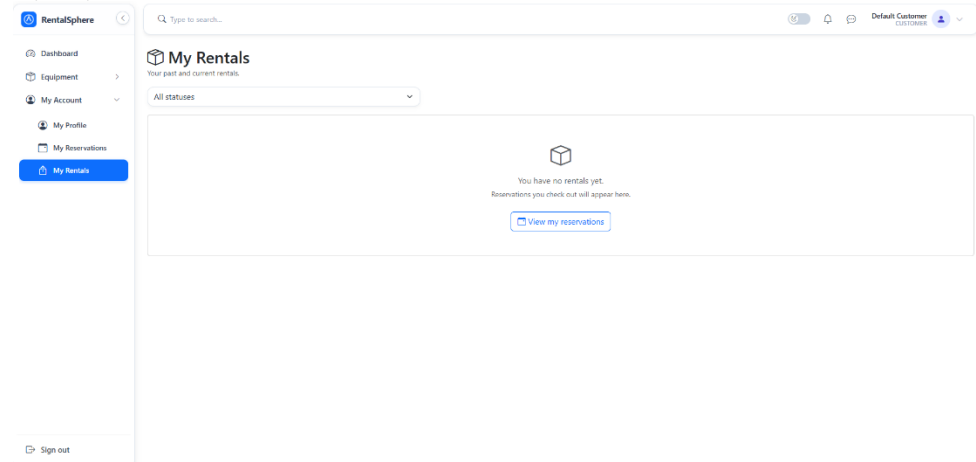
Customer My Profile

The signed-in customer sees a read-only page with their personal details (name, e-mail, phone, address), current loyalty tier and total spent, plus a link to My Reservations. No edit or delete actions are available; only staff/admin can modify the record.



My Reservations (Customer)

After clicking “My Reservations” the customer sees a card grid that lists only their own bookings. Each card shows the reservation number, status badge (Pending, Confirmed, Checked-Out, Completed,

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>Cancelled), rental dates, number of items, and estimated total cost. A View details button opens a read-only page with the full line items and assigned serial numbers. From the grid or details page the customer can Create a new reservation (which opens the reservation form) and Cancel a reservation only while it is still Pending; later-stage reservations cannot be cancelled by the customer. No edit, confirm, or checkout actions appear.</p>                                                                                                                                     |
|         | <p><b>My Rentals (Customer)</b></p> <p>A read-only list that shows every rental transaction belonging to the signed-in customer. Each entry displays the transaction number, the equipment rented (with serial numbers), the checkout and return dates, the final status (Active, Completed, Overdue), and the linked invoice balance. From this page the customer can click View details to see the full timeline, line items, condition notes, and the invoice with a “Print PDF” button. No actions to create, edit, or cancel rentals are available – the page is purely informational.</p>  |
| Backend | <p><b>Program.cs – DI &amp; hosted worker (Program.cs 12-45)</b></p> <p>Bootstraps EF Core, Identity, cookie auth, registers every module’s repo/service, adds the ReservationExpiryWorker background service and the global exception filter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> var builder = WebApplication.CreateBuilder(args); QuestPDF.Settings.License = LicenseType.Community;  var cs = builder.Configuration.GetConnectionString("DefaultConnection") ?? throw new InvalidOperationException("Missing DefaultConnection");  builder.Services.AddDbContext&lt;ApplicationDbContext&gt;(o =&gt;     o.UseSqlServer(cs).ConfigureWarnings(w =&gt; w.Ignore(         Microsoft.EntityFrameworkCore.Diagnostics.RelationalEventId.PendingModelChangesWarning));  builder.Services.AddIdentity&lt;ApplicationUser, IdentityRole&gt;(o =&gt; {     o.Password.RequireDigit = true;     o.Password.RequiredLength = 8;     o.User.RequireUniqueEmail = true; }).AddEntityFrameworkStores&lt;ApplicationDbContext&gt;() .AddDefaultTokenProviders();  builder.Services.ConfigureApplicationCookie(o =&gt; {     o.LoginPath = "/Account/Login";     o.AccessDeniedPath = "/Account/AccessDenied";     o.ExpireTimeSpan = TimeSpan.FromHours(8); });  builder.Services.AddHttpContextAccessor(); builder.Services.AddScoped&lt;ICurrentUser, CurrentUser&gt;(); builder.Services.AddScoped&lt;IAuditLogger, AuditLogger&gt;();  // ...module registrations (Equipment, Customer, Reservation, Billing...)...  builder.Services.AddSingleton(new ReservationExpiryOptions {     PendingExpiryHours = 0.5,     ConfirmedExpiryDays = 1,     ScanIntervalMinutes = 5 }); builder.Services.AddHostedService&lt;ReservationExpiryWorker&gt;();  builder.Services.AddControllersWithViews(o =&gt; o.Filters.Add&lt;GlobalExceptionHandler&gt;());  var app = builder.Build(); app.UseAuthentication(); app.UseAuthorization(); app.MapControllerRoute("default", "{controller=Home}/{action=Index}/{id?}"); app.Run(); </pre> |
|  | <p><b>ApplicationDbContext.cs – Core model (ApplicationDbContext.cs 21-55)</b><br/> Declares every DbSet for the ten modules and merges ASP.NET Core Identity via IdentityDbContext. The Fluent-API in OnModelCreating (omitted here) defines keys, relationships, enum-to-string conversions and indexes – the single source of truth for the schema.</p> <pre> public class ApplicationDbContext : IdentityDbContext&lt;ApplicationUser&gt; {     public ApplicationDbContext(DbContextOptions&lt;ApplicationDbContext&gt; o) : base(o) { }      public DbSet&lt;Category&gt; Categories =&gt; Set&lt;Category&gt;();     public DbSet&lt;EquipmentCatalog&gt; Equipment =&gt; Set&lt;EquipmentCatalog&gt;();     public DbSet&lt;EquipmentItem&gt; EquipmentItems =&gt; Set&lt;EquipmentItem&gt;();     public DbSet&lt;Customer&gt; Customers =&gt; Set&lt;Customer&gt;();     public DbSet&lt;Reservation&gt; Reservations =&gt; Set&lt;Reservation&gt;();     public DbSet&lt;ReservationItem&gt; ReservationItems =&gt; Set&lt;ReservationItem&gt;();     public DbSet&lt;RentalTransaction&gt; RentalTransactions =&gt; Set&lt;RentalTransaction&gt;();     public DbSet&lt;Invoice&gt; Invoices =&gt; Set&lt;Invoice&gt;();     public DbSet&lt;InvoiceLine&gt; InvoiceLines =&gt; Set&lt;InvoiceLine&gt;();     public DbSet&lt;Payment&gt; Payments =&gt; Set&lt;Payment&gt;();     public DbSet&lt;DamagePenalty&gt; DamagePenalties =&gt; Set&lt;DamagePenalty&gt;();     public DbSet&lt;AuditLog&gt; AuditLogs =&gt; Set&lt;AuditLog&gt;();     // CRM, Maintenance omitted for brevity } </pre>                                                                                                                             |
|  | <p><b>ReservationService.cs – Availability check (ReservationService.cs 214-240)</b><br/> Half-open day-interval overlap algorithm that runs entirely in SQL (SumAsync). Returns per-line Available count and a boolean IsAvailable used by both create- and confirm-flows and by the UI calendar.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> public async Task&lt;AvailabilityResultDto&gt; CheckAvailabilityAsync(     DateTime start, DateTime end,     List&lt;ReservationItemInputDto&gt; items,     int? excludeReservationId = null) {     var result = new AvailabilityResultDto { IsAvailable = true };      foreach (var line in items)     {         var eq = await _equipment.GetAsync(line.EquipmentID)             ?? throw new NotFoundException(\$"Equipment {line.EquipmentID} not found.");          var totalStock = eq.StockQuantity &gt; 0 ? eq.StockQuantity             : await _items.CountAsyncByEquipment(line.EquipmentID,                 AvailabilityStatus.Available);          var conflicted = await _db.ReservationItems             .Where(ri =&gt; ri.EquipmentID == line.EquipmentID                 &amp;&amp; (ri.Reservation.Status == ReservationStatus.Pending                        ri.Reservation.Status == ReservationStatus.Confirmed                        ri.Reservation.Status == ReservationStatus.CheckedOut)                 &amp;&amp; (excludeReservationId == null    ri.ReservationID != excludeReservationId)                 &amp;&amp; ri.Reservation.RentalStartDate.Date &lt; end.Date                 &amp;&amp; ri.Reservation.RentalEndDate.Date &gt; start.Date)             .SumAsync(ri =&gt; (int?)ri.Quantity) ?? 0;          var available = Math.Max(0, totalStock - conflicted);          result.Lines.Add(new AvailabilityLineResultDto         {             EquipmentID = line.EquipmentID,             EquipmentName = eq.Name,             Requested = line.Quantity,             Available = available,             IsAvailable = available &gt;= line.Quantity,             Reason = available &gt;= line.Quantity ? null                 : \$"Only {available} unit(s) free {start:d}-{end:d}; requested {line.Quantity}."         });         if (!result.Lines.Last().IsAvailable) result.IsAvailable = false;     }     return result; } </pre> |
|  | <p><b>ReservationsController.cs – RBAC-protected actions (ReservationsController.cs 15-70)</b></p> <p>Shows the three public entry points: staff/admin list (Index), customer-only list (MyReservations), and the create flow with automatic customer-scope enforcement. Confirm is limited to Admin/Staff via [Authorize(Roles=...)].</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

```

[Authorize]
public class ReservationsController : Controller
{
    private readonly IReservationService _svc;
    private readonly ICustomerService _cust;
    private readonly IEquipmentService _eq;
    private readonly ICurrentUser _cu;

    public ReservationsController(IReservationService s, ICustomerService c,
                                IEquipmentService e, ICurrentUser u)
    { _svc=s; _cust=c; _eq=e; _cu=u; }

    // Staff / Admin - paged, searchable, sortable
    public async Task<IActionResult> Index(string? q, string? f, string? s,
                                           int p=1, int ps=20)
    {
        var (items, tot) = await _svc.ListPagedAsync(q, f, s, (p-1)*ps, ps);
        return View(new ReservationIndexVM{ Items=items, Search=q, StatusFilter=f,
                                             Sort=s, Page=p, PageSize=ps, TotalCount=tot});
    }

    // Customer - only own reservations
    [Authorize(Roles = RoleNames.Customer)]
    public async Task<IActionResult> MyReservations(string? f, int p=1, int ps=20)
    {
        var (items, tot) = await _svc.ListMinePagedAsync(f, (p-1)*ps, ps);
        return View(new MyReservationsVM{
            Items=items, StatusFilter=f, Page=p, PageSize=ps, TotalCount=tot,
            PendingCount=await _svc.CountMineByStatusAsync(nameof(ReservationStatus.Pending)),
            ConfirmedCount=await _svc.CountMineByStatusAsync(nameof(ReservationStatus.Confirmed))});
    }

    // Create - GET
    [HttpGet]
    public async Task<IActionResult> Create() => View(new ReservationCreateVM{
        Form=new ReservationCreateDto{
            RentalStartDate=DateTime.UtcNow.Date.AddDays(1),
            RentalEndDate=DateTime.UtcNow.Date.AddDays(2)},
        Customers=await _cust.GetLookupAsync(),
        Equipment=await _eq.GetLookupAsync()});

    // Customer - only own reservations
    [Authorize(Roles = RoleNames.Customer)]
    public async Task<IActionResult> MyReservations(string? f, int p=1, int ps=20)
    {
        var (items, tot) = await _svc.ListMinePagedAsync(f, (p-1)*ps, ps);
        return View(new MyReservationsVM{
            Items=items, StatusFilter=f, Page=p, PageSize=ps, TotalCount=tot,
            PendingCount=await _svc.CountMineByStatusAsync(nameof(ReservationStatus.Pending)),
            ConfirmedCount=await _svc.CountMineByStatusAsync(nameof(ReservationStatus.Confirmed))});
    }

    // Create - GET
    [HttpGet]
    public async Task<IActionResult> Create() => View(new ReservationCreateVM{
        Form=new ReservationCreateDto{
            RentalStartDate=DateTime.UtcNow.Date.AddDays(1),
            RentalEndDate=DateTime.UtcNow.Date.AddDays(2)},
        Customers=await _cust.GetLookupAsync(),
        Equipment=await _eq.GetLookupAsync()});

    // Create - POST (customer-scope enforcement)
    [HttpPost, ValidateAntiForgeryToken]
    public async Task<IActionResult> Create(ReservationCreateVM vm)
    {
        if (!ModelState.IsValid) { vm.Customers=await _cust.GetLookupAsync(); vm.Equipment=await _eq.GetLookupAsync(); return View(vm); }
        if (!_cu.IsCustomerScoped)
        {
            var myId=await _cust.GetMyCustomerIdAsync();
            if (myId is null){ ModelState.AddModelError("", "No linked customer."); return View(vm); }
            vm.Form.CustomerID=myId.Value;
        }
        var id=await _svc.CreateAsync(vm.Form, _cu.UserId?"system");
        TempData["Success"]+="Reservation #{id} created.";
        return RedirectToAction(nameof(Details), new{id});
    }

    // Confirm - Admin/Staff only
    [Authorize(Roles = $"{RoleNames.Admin},{RoleNames.Staff}")]
    [HttpPost, ValidateAntiForgeryToken]
    public async Task<IActionResult> Confirm(int id)
    {
        try{ await _svc.ConfirmAsync(id, _cu.UserId?"system"); TempData["Success"]+="Reservation #{id} confirmed."; }
        catch (InvalidOperationException ex){ TempData["Error"]=ex.Message; }
        return RedirectToAction(nameof(Details), new{id});
    }
}

```

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <p><b>AvailabilityService.cs – Calendar query (AvailabilityService.cs 1-28)</b></p> <p>Read-only service used by the front-end date picker. One EF query pulls all overlapping reservation lines, then a tiny in-memory loop builds a day-by-day Free/Booked matrix – identical overlap logic to CheckAvailabilityAsync.</p> <pre> public async Task&lt;AvailabilityCalendarDto&gt; GetCalendarAsync(int equipmentId,     DateTime start, DateTime end) {     var eq = await _equipment.GetAsync(equipmentId)         ?? throw new NotFoundException(\$"Equipment {equipmentId} not found.");      var total = eq.StockQuantity &gt; 0 ? eq.StockQuantity         : await _items.CountAsyncByEquipment(equipmentId, AvailabilityStatus.Available);      var overlaps = await _db.ReservationItems         .Where(ri =&gt; ri.EquipmentID == equipmentId             &amp;&amp; (ri.Reservation.Status == ReservationStatus.Pending                    ri.Reservation.Status == ReservationStatus.Confirmed                    ri.Reservation.Status == ReservationStatus.CheckedOut)             &amp;&amp; ri.Reservation.RentalStartDate.Date &lt; end.Date             &amp;&amp; ri.Reservation.RentalEndDate.Date &gt; start.Date)         .Select(ri =&gt; new { ri.Reservation.RentalStartDate,             ri.Reservation.RentalEndDate,             ri.Quantity })         .ToListAsync();      var days = new List&lt;DayAvailabilityDto&gt;();     for (var d = start.Date; d &lt; end.Date; d = d.AddDays(1))     {         var booked = overlaps.Where(o =&gt; o.RentalStartDate.Date &lt;= d &amp;&amp; o.RentalEndDate.Date &gt; d)             .Sum(o =&gt; o.Quantity);         days.Add(new DayAvailabilityDto(Date=d, TotalStock=total, Booked=booked, Free=Math.Max(0, total-booked)));     }     return new AvailabilityCalendarDto{EquipmentID=equipmentId, EquipmentName=eq.Name, Days=days}; } </pre> |
| <p><b>API FUNCTIONS/<br/>FEATURES</b></p> | <p><b>EquipmentController.cs – Create equipment (EquipmentController.cs 30-45)</b></p> <p>Admin/Staff post a new catalogue item; the service validates, persists and logs the change.</p> <pre> [Authorize(Roles = \$"{RoleNames.Admin},{RoleNames.Staff}")] [HttpPost, ValidateAntiForgeryToken] public async Task&lt;ActionResult&gt; Create(EquipmentCreateVM vm) {     if (!ModelState.IsValid) return View(vm);      var id = await _eqSvc.CreateAsync(vm.Form, User.GetUserId());     TempData["Success"] = \$"Equipment #{id} added.";     return RedirectToAction(nameof(Details), new { id }); } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                           | <p><b>CustomersController.cs – Edit customer (CustomersController.cs 28-44)</b></p> <p>Staff/Admin can edit any profile; a customer may only edit their own linked record (enforced in service).</p> <pre> [HttpPost, ValidateAntiForgeryToken] public async Task&lt;ActionResult&gt; Edit(int id, CustomerEditVM vm) {     if (id != vm.Form.CustomerID) return BadRequest();     if (!ModelState.IsValid) return View(vm);      try     {         await _custSvc.UpdateAsync(vm.Form, User.GetUserId());         TempData["Success"] = "Customer updated.";         return RedirectToAction(nameof(Details), new { id });     }     catch (ForbiddenException ex)     {         ModelState.AddModelError("", ex.Message);         return View(vm);     } } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### ReservationService.cs – Create reservation (ReservationService.cs 268-340)

Validates dates, runs the half-open availability check, calculates total, persists a Pending reservation and writes an audit entry.

```
public async Task<int> CreateAsync(ReservationCreatedDto dto, string actorId)
{
    if (dto.RentalEndDate <= dto.RentalStartDate)
        throw new InvalidOperationException("End date must be after start date.");
    if (dto.Items?.Count == 0)
        throw new InvalidOperationException("At least one equipment line required.");

    var avail = await CheckAvailabilityAsync(dto.RentalStartDate, dto.RentalEndDate, dto.Items);
    if (!avail.IsAvailable)
        throw new InvalidOperationException(avail.Lines.First(l => !l.IsAvailable).Reason);

    var days = Math.Max(1, (int)Math.Ceiling((dto.RentalEndDate.Date - dto.RentalStartDate.Date).TotalDays));
    decimal total = 0m;
    var lines = new List<ReservationItem>();

    foreach (var i in dto.Items)
    {
        var eq = await _equipment.GetAsync(i.EquipmentID)
            ?? throw new NotFoundException($"Equipment {i.EquipmentID} not found.");
        total += eq.DailyRate * i.Quantity * days;
        lines.Add(new ReservationItem { EquipmentID = eq.EquipmentID, Quantity = i.Quantity, Notes = i.Notes });
    }

    var r = new Reservation
    {
        CustomerID = dto.CustomerID,
        ReservationDate = DateTime.UtcNow,
        RentalStartDate = dto.RentalStartDate,
        RentalEndDate = dto.RentalEndDate,
        Status = ReservationStatus.Pending,
        TotalEstimatedCost = total,
        Notes = dto.Notes,
        CreatedByUserId = _current.UserId,
        Items = lines
    };

    await _repo.AddAsync(r);
    await _repo.SaveChangesAsync();

    await _audit.LogAsync(actorId, "Reservation.Create", "Reservation", r.ReservationID.ToString(),
        newValues: new { r.CustomerID, r.RentalStartDate, r.RentalEndDate, Items = lines.Count, r.TotalEstimatedCost });

    return r.ReservationID;
}
```

### ReservationService.cs – Confirm reservation (ReservationService.cs 342-417)

Staff/Admin only; re-checks availability, pins concrete EquipmentItems, flips their status to Reserved, updates reservation to Confirmed, audits.

```
public async Task ConfirmAsync(int id, string actorId)
{
    if (!_current.IsAdmin && !_current.IsStaff)
        throw new ForbiddenException("Only Admin or Staff can confirm reservations.");

    var r = await _repo.GetWithItemsAsync(id) ?? throw new NotFoundException($"Reservation {id} not found.");
    if (r.Status != ReservationStatus.Pending)
        throw new InvalidOperationException($"Only Pending reservations can be confirmed (current: {r.Status}).");

    // re-check availability excluding this reservation
    var inputs = r.Items.Select(i => new ReservationItemInputDto { EquipmentID = i.EquipmentID, Quantity = i.Quantity }).ToList();
    var avail = await CheckAvailabilityAsync(r.RentalStartDate, r.RentalEndDate, inputs, excludeReservationId: id);
    if (!avail.IsAvailable)
        throw new InvalidOperationException(avail.Lines.First(l => !l.IsAvailable).Reason);

    // assign concrete items
    var assigned = new Dictionary<int, List<int>>();
    foreach (var line in r.Items)
    {
        var free = await _items.ListFreeForRangeAsync(line.EquipmentID, r.RentalStartDate, r.RentalEndDate, excludeReservationId: id);
        var picked = free.Take(line.Quantity).Select(x => x.ItemID).ToList();
        if (picked.Count < line.Quantity) throw new InvalidOperationException("Not enough free units.");
        assigned[line.EquipmentID] = picked;
        line.AssignedItemID = picked.First();
    }

    // mark items Reserved
    foreach (var itemId in assigned.Values.SelectMany(x => x).Distinct())
        await _itemService.ChangeStatusAsync(itemId, nameof(AvailabilityStatus.Reserved), actorId);

    r.Status = ReservationStatus.Confirmed;
    _repo.Update(r);
    await _repo.SaveChangesAsync();

    await _audit.LogAsync(actorId, "Reservation.Confirm", "Reservation", id.ToString(),
        oldValues: new { Status = ReservationStatus.Pending.ToString() },
        newValues: new { Status = ReservationStatus.Confirmed.ToString(), AssignedItems = assigned.Values.SelectMany(x => x) });
}
```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>ReservationService.cs – Cancel reservation (ReservationService.cs 419-512)</b></p> <p>Owner may cancel only own Pending reservation; Staff/Admin can cancel any non-terminal state. Releases assigned items back to Available when confirmed.</p> <pre> public async Task CancelAsync(int id, string actorId, ReservationCancellationReason reason, string? notes) {     var r = await _repo.GetWithItemsAsync(id) ?? throw new NotFoundException(\$"Reservation {id} not found.");      if (_current.IsCustomerScoped)     {         if (r.Customer.UserId != _current.UserId)             throw new ForbiddenException("You can only cancel your own reservations.");         if (r.Status != ReservationStatus.Pending)             throw new InvalidOperationException("Only pending reservations can be cancelled by customers.");     }     else if (!_current.IsAdmin &amp;&amp; !_current.IsStaff)         throw new ForbiddenException("Only Admin or Staff can cancel reservations.");      if (r.Status is ReservationStatus.Cancelled or ReservationStatus.Expired) return;      // release items if confirmed     if (r.Status == ReservationStatus.Confirmed)     {         var equipIds = r.Items.Select(i =&gt; i.EquipmentID).Distinct().ToList();         var reserved = await _db.EquipmentItems             .Where(ei =&gt; equipIds.Contains(ei.EquipmentID) &amp;&amp; ei.AvailabilityStatus == AvailabilityStatus.Reserved)             .Select(ei =&gt; ei.ItemID).ToListAsync();          var explicitIds = r.Items.Where(i =&gt; i.AssignedItemID.HasValue).Select(i =&gt; i.AssignedItemID!.Value).Distinct().ToList();         var toRelease = new HashSet&lt;int&gt;(explicitIds);         foreach (var line in r.Items)         {             var extra = reserved.Where(x =&gt; !toRelease.Contains(x)).Take(line.Quantity).ToList();             foreach (var e in extra) toRelease.Add(e);         }         foreach (var itemID in toRelease)             await _itemService.ChangeStatusAsync(itemID, nameof(AvailabilityStatus.Available), actorId);     }      r.Status = ReservationStatus.Cancelled;     r.CancellationReason = reason;     r.CancellationNotes = string.IsNullOrEmpty(notes) ? null : notes.Trim();     r.CancelledAt = DateTime.UtcNow;     _repo.Update(r);     await _repo.SaveChangesAsync();      await _audit.LogAsync(actorId, "Reservation.Cancel", "Reservation", id.ToString(),         oldValues: new { Status = r.Status.ToString() },         newValues: new { Status = ReservationStatus.Cancelled.ToString(), CancellationReason = reason.ToString(), HashNotes = !string.IsNullOrEmpty(notes) }); } </pre> |
|  | <p><b>RentalTransactionService.cs – Checkout (RentalTransactionService.cs 25-60)</b></p> <p>Staff/Admin creates a RentalTransaction from a confirmed reservation, copies lines, sets CheckoutDate, marks items CheckedOut, and audits.</p> <pre> public async Task&lt;int&gt; CheckoutAsync(int reservationId, string actorId) {     var res = await _resRepo.GetWithItemsAsync(reservationId)         ?? throw new NotFoundException(\$"Reservation {reservationId} not found.");     if (res.Status != ReservationStatus.Confirmed)         throw new InvalidOperationException("Only Confirmed reservations can be checked out.");      var txn = new RentalTransaction     {         ReservationID = res.ReservationID,         CustomerID = res.CustomerID,         CheckoutDate = DateTime.UtcNow,         Status = RentalTransactionStatus.Active,         ProcessedByUserID = _current.UserId,         Items = res.Items.Select(i =&gt; new RentalTransactionItem         {             EquipmentID = i.EquipmentID,             EquipmentItemID = i.AssignedItemID,             Quantity = i.Quantity,             DailyRate = i.Equipment?.DailyRate ?? 0m         }).ToList()     };      // flip items to CheckedOut     foreach (var item in txn.Items.Where(i =&gt; i.EquipmentItemID.HasValue))         await _itemSvc.ChangeStatusAsync(item.EquipmentItemID!.Value, nameof(AvailabilityStatus.CheckedOut), actorId);      await _txnRepo.AddAsync(txn);     await _txnRepo.SaveChangesAsync();      res.Status = ReservationStatus.CheckedOut;     _resRepo.Update(res);     await _resRepo.SaveChangesAsync();      await _audit.LogAsync(actorId, "RentalTransaction.Checkout", "RentalTransaction", txn.RentalTransactionID.ToString(),         newValues: new { txn.ReservationID, txn.CustomerID, txn.CheckoutDate, ItemCount = txn.Items.Count });      return txn.RentalTransactionID; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | <p><b>RentalTransactionService.cs – Return / Check-in (RentalTransactionService.cs 62-95)</b></p> <p>Staff records return date, condition notes, flips items to Available (or Maintenance if flagged), updates transaction to Completed, creates invoice via BillingService.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

```

public async Task ReturnAsync(int txnId, DateTime returnDate, string? conditionNotes, string actorId)
{
    var txn = await _txnRepo.GetWithItemsAsync(txnId)
        ?? throw new NotFoundException($"Transaction {txnId} not found.");
    if (txn.Status != RentalTransactionStatus.Active)
        throw new InvalidOperationException("Only active rentals can be returned.");

    txn.ReturnDate = returnDate;
    txn.ConditionNotes = conditionNotes;
    txn.Status = RentalTransactionStatus.Completed;
    txn.ReturnedToUserId = _current.UserId;

    foreach (var line in txn.Items.Where(i => i.EquipmentItemID.HasValue))
    {
        var targetStatus = line.Damaged ? nameof(AvailabilityStatus.Maintenance) : nameof(AvailabilityStatus.Available);
        await _itemSvc.ChangeStatusAsync(line.EquipmentItemID!.Value, targetStatus, actorId);
    }

    _txnRepo.Update(txn);
    await _txnRepo.SaveChangesAsync();

    // auto-generate invoice
    await _billingSvc.GenerateInvoiceAsync(txn.RentalTransactionID, actorId);

    await _audit.LogAsync(actorId, "RentalTransaction.Return", "RentalTransaction", txnId.ToString(),
        newValues: new { txn.ReturnDate, txn.ConditionNotes, txn.Status });
}

```

### BillingService.cs – Generate invoice (BillingService.cs 18-48)

Builds an invoice from a completed rental transaction: rental lines, late-fee lines (if any), damage-penalty lines, persists and logs.

```

public async Task<int> GenerateInvoiceAsync(int rentalTxnId, string actorId)
{
    var txn = await _txnRepo.GetWithItemsAsync(rentalTxnId)
        ?? throw new NotFoundException($"RentalTransaction {rentalTxnId} not found.");
    if (txn.Status != RentalTransactionStatus.Completed)
        throw new InvalidOperationException("Invoice can only be generated for completed rentals.");

    var invoice = new Invoice
    {
        RentalTransactionID = txn.RentalTransactionID,
        CustomerID = txn.CustomerID,
        InvoiceNumber = $"INV-{DateTime.UtcNow:yyyyMMdd}-{txn.RentalTransactionID:D6}",
        InvoiceDate = DateTime.UtcNow,
        DueDate = DateTime.UtcNow.AddDays(_options.PaymentTermsDays),
        Status = InvoiceStatus.Open,
        CreatedByUserId = _current.UserId,
        Lines = new List<InvoiceLine>()
    };

    // rental lines
    foreach (var l in txn.Items)
    {
        var days = Math.Max(1, (int)Math.Ceiling((txn.ReturnDate!.Value.Date - txn.CheckoutDate.Date).TotalDays));
        var lineTotal = l.DailyRate * l.Quantity * days;
        invoice.Lines.Add(new InvoiceLine
        {
            LineType = InvoiceLineType.Rental,
            Description = $"{l.Equipment?.Name} x{l.Quantity} ({days} day(s))",
            Quantity = l.Quantity,
            UnitAmount = l.DailyRate,
            LineTotal = lineTotal
        });
        invoice.SubTotal += lineTotal;
    }

    // late fees
    var lateDays = (txn.ReturnDate!.Value.Date - txn.RentalEndDate.Date).Days;
    if (lateDays > 0)
    {
        var fee = _options.LateFeePerDay * lateDays;
        invoice.Lines.Add(new InvoiceLine
        {
            LineType = InvoiceLineType.LateFee,
            Description = $"Late return {lateDays} day(s) @ {_options.LateFeePerDay:C}/day",
            Quantity = lateDays,
            UnitAmount = _options.LateFeePerDay,

```

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|  | <pre> // Late fees var lateDays = (txn.ReturnDate!.Value.Date - txn.RentalEndDate.Date).Days; if (lateDays &gt; 0) {     var fee = _options.LateFeePerDay * lateDays;     invoice.Lines.Add(new InvoiceLine     {         LineType = InvoiceLineType.LateFee,         Description = \$"Late return {lateDays} day(s) @ {_options.LateFeePerDay:C}/day",         Quantity = lateDays,         UnitAmount = _options.LateFeePerDay,         LineTotal = fee     });     invoice.SubTotal += fee; }  // damage penalties (un-waived) var penalties = await _dmgRepo.ListUnwaivedByTransactionAsync(txn.RentalTransactionID); foreach (var p in penalties) {     invoice.Lines.Add(new InvoiceLine     {         LineType = InvoiceLineType.DamagePenalty,         Description = p.Description,         Quantity = 1,         UnitAmount = p.Amount - p.WaivedAmount,         LineTotal = p.Amount - p.WaivedAmount     });     invoice.SubTotal += p.Amount - p.WaivedAmount; }  invoice.TotalAmount = invoice.SubTotal; await _billRepo.AddAsync(invoice); await _billRepo.SaveChangesAsync();  await _audit.LogAsync(actorId, "Invoice.Generate", "Invoice", invoice.InvoiceID.ToString(),     newValues: new { invoice.InvoiceNumber, invoice.TotalAmount, invoice.DueDate });  return invoice.InvoiceID; } </pre>                                                                                                  |
|  | <p><b>DamagePenaltyService.cs – Assess damage (DamagePenaltyService.cs 22-55)</b></p> <p>Staff records a damage/penalty against a returned transaction; amount auto-added to next invoice unless waived by Admin.</p> <pre> public async Task&lt;int&gt; AssessAsync(DamagePenaltyCreateDto dto, string actorId) {     var txn = await _txnRepo.GetAsync(dto.RentalTransactionID)         ?? throw new NotFoundException(\$"Transaction {dto.RentalTransactionID} not found.");     if (txn.Status != RentalTransactionStatus.Completed)         throw new InvalidOperationException("Penalties only allowed on completed rentals.");      var penalty = new DamagePenalty     {         RentalTransactionID = txn.RentalTransactionID,         EquipmentID = dto.EquipmentID,         EquipmentItemID = dto.EquipmentItemID,         PenaltyType = dto.PenaltyType,         Description = dto.Description,         Amount = dto.Amount,         Status = DamagePenaltyStatus.Pending,         ReportedByUserId = _current.UserId,         ReportedAt = DateTime.UtcNow     };      await _repo.AddAsync(penalty);     await _repo.SaveChangesAsync();      await _audit.LogAsync(actorId, "DamagePenalty.Assess", "DamagePenalty", penalty.DamagePenaltyID.ToString(),         newValues: new { penalty.RentalTransactionID, penalty.PenaltyType, penalty.Amount });      return penalty.DamagePenaltyID; } </pre> |
|  | <p><b>MaintenanceService.cs – Open maintenance record (MaintenanceService.cs 20-48)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                                 | <p>Staff creates a MaintenanceRecord for an EquipmentItem, sets status InProgress, item status becomes Maintenance; only Admin can close/delete</p> <pre> public async Task&lt;int&gt; OpenAsync(MaintenanceCreateDto dto, string actorId) {     var item = await _itemRepo.GetAsync(dto.EquipmentItemID)         ?? throw new NotFoundException(\$"EquipmentItem {dto.EquipmentItemID} not found.");      var rec = new MaintenanceRecord     {         EquipmentItemID = item.ItemID,         Reason = dto.Reason,         Notes = dto.Notes,         StartedAt = DateTime.UtcNow,         Status = MaintenanceStatus.InProgress,         OpenedByUserId = _current.UserId     };      await _repo.AddAsync(rec);     await _itemSvc.ChangeStatusAsync(item.ItemID, nameof(AvailabilityStatus.Maintenance), actorId);     await _repo.SaveChangesAsync();      await _audit.LogAsync(actorId, "Maintenance.Open", "MaintenanceRecord", rec.MaintenanceRecordID.ToString(),         newValues: new { rec.EquipmentItemID, rec.Reason, rec.StartedAt });      return rec.MaintenanceRecordID; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                 | <p><b>ReportsService.cs – Revenue report (ReportsService.cs 30-58)</b><br/>Generates a date-range revenue summary (rental + late fees + penalties) grouped by equipment category; returns DTO for PDF export (QuestPDF).</p> <pre> public async Task&lt;RevenueReportDto&gt; RevenueAsync(DateTime from, DateTime to) {     var invoices = await _billRepo.Query()         .Where(i =&gt; i.InvoiceDate &gt;= from &amp;&amp; i.InvoiceDate &lt;= to &amp;&amp; i.Status != InvoiceStatus.Voided)         .Include(i =&gt; i.Lines)         .ToListAsync();      var rental = invoices.SelectMany(i =&gt; i.Lines)         .Where(l =&gt; l.LineType == InvoiceLineType.Rental)         .Sum(l =&gt; l.LineTotal);      var late = invoices.SelectMany(i =&gt; i.Lines)         .Where(l =&gt; l.LineType == InvoiceLineType.LateFee)         .Sum(l =&gt; l.LineTotal);      var penalties = invoices.SelectMany(i =&gt; i.Lines)         .Where(l =&gt; l.LineType == InvoiceLineType.DamagePenalty)         .Sum(l =&gt; l.LineTotal);      var byCategory = await _db.InvoiceLines         .Where(l =&gt; l.Invoice.InvoiceDate &gt;= from &amp;&amp; l.Invoice.InvoiceDate &lt;= to             &amp;&amp; l.LineType == InvoiceLineType.Rental)         .Join(_db.Equipment, l =&gt; l.Description, e =&gt; e.Name, (l, e) =&gt; new { l.LineTotal, e.CategoryID })         .GroupBy(x =&gt; x.CategoryID)         .Select(g =&gt; new CategoryRevenueDto         {             CategoryID = g.Key,             CategoryName = _db.Categories.First(c =&gt; c.CategoryID == g.Key).Name,             Revenue = g.Sum(x =&gt; x.LineTotal)         })         .ToListAsync();      return new RevenueReportDto     {         From = from,         To = to,         RentalRevenue = rental,         LateFeeRevenue = late,         PenaltyRevenue = penalties,         TotalRevenue = rental + late + penalties,         ByCategory = byCategory     }; } </pre> |
| <p><b>SECURITY FEATURES</b></p> | <p><b>Program.cs – Authentication &amp; cookie hardening (Program.cs 42-50)</b><br/>Configures the Identity cookie with a short sliding expiration, secure-policy, and explicit login/access-denied paths.</p> <pre> builder.Services.ConfigureApplicationCookie(o =&gt; {     o.LoginPath = "/Account/Login";     o.AccessDeniedPath = "/Account/AccessDenied";     o.ExpireTimeSpan = TimeSpan.FromHours(8);     o.SlidingExpiration = true;     o.Cookie.SecurePolicy = CookieSecurePolicy.SameAsRequest; // HTTPS only in prod     o.Cookie.HttpOnly = true; // JS cannot read     o.Cookie.SameSite = SameSiteMode.Strict; // CSRF mitigation }); </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|  | <p><b>Program.cs – Data-Protection key persistence (Program.cs 129-132)</b><br/>Stores encryption keys on the file system so they survive container restarts and work in a web-farm.</p> <pre>builder.Services.AddDataProtection()     .PersistKeysToFileSystem(new DirectoryInfo(@"C:\Home\DataProtectionKeys"))     .SetApplicationName("RentalSphere");</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|  | <p><b>Common/Constants/RoleNames.cs – Centralised role constants (RoleNames.cs 1-9)</b><br/>Single source of truth for role strings; prevents typos in [Authorize] and service checks.</p> <pre>namespace RentalSphere.Common.Constants;  public static class RoleNames {     public const string Admin = "Admin";     public const string Staff = "Staff";     public const string Customer = "Customer"; }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <p><b>Common/Services/CurrentUser.cs – Runtime principal wrapper (CurrentUser.cs 12-30)</b><br/>Exposes typed IsAdmin, IsStaff, IsCustomerScoped properties used by every service/controller for fine-grained RBAC.</p> <pre>public class CurrentUser : ICurrentUser {     private readonly IHttpContextAccessor _http;     public CurrentUser(IHttpContextAccessor http) =&gt; _http = http;      public string? UserId =&gt; _http.HttpContext?.User?.FindFirstValue(ClaimTypes.NameIdentifier);     public string? Email =&gt; _http.HttpContext?.User?.FindFirstValue(ClaimTypes.Email);      public bool IsAdmin =&gt; _http.HttpContext?.User?.IsInRole(RoleNames.Admin) ?? false;     public bool IsStaff =&gt; _http.HttpContext?.User?.IsInRole(RoleNames.Staff) ?? false;     public bool IsCustomerScoped =&gt; _http.HttpContext?.User?.IsInRole(RoleNames.Customer) ?? false; }</pre> |
|  | <p><b>ReservationService.cs – Role-guarded Confirm (ReservationService.cs 342-347)</b><br/>Service-level guard ensures only Admin or Staff can confirm, even if a malicious request bypasses the controller attribute.</p> <pre>public async Task ConfirmAsync(int id, string actorId) {     if (!_current.IsAdmin &amp;&amp; !_current.IsStaff)         throw new ForbiddenException("Only Admin or Staff can confirm reservations.");      var r = await _repo.GetWithItemsAsync(id)         ?? throw new NotFoundException(\$"Reservation {id} not found.");     // ... rest of confirm logic ... }</pre>                                                                                                                                                                                                                                                                                       |
|  | <p><b>ReservationService.cs – Customer-scope Create (ReservationService.cs 279-284)</b><br/>When a Customer creates a reservation the service forces the CustomerID to the logged-in user's linked customer record.</p> <pre>if (_current.IsCustomerScoped) {     if (string.IsNullOrEmpty(customer.UserId)    customer.UserId != _current.UserId)         throw new ForbiddenException("You can only create reservations for yourself."); }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <p><b>Identity/Services/AuditLogger.cs – Immutable audit trail (AuditLogger.cs 10-27)</b><br/>Every state-changing operation writes a tamper-evident row (actor, action, entity, before/after JSON, IP).</p> <pre>public async Task LogAsync(string actorUserId, string action, string entityName,     string entityId, object? oldValues = null,     object? newValues = null, string? ip = null) {     var entry = new AuditLog     {         ActorUserId = actorUserId,         Action = action,         EntityName = entityName,         EntityId = entityId,         OldValuesJson = oldValues == null ? null : JsonSerializer.Serialize(oldValues),         NewValuesJson = newValues == null ? null : JsonSerializer.Serialize(newValues),         IPAddress = ip,         Timestamp = DateTime.UtcNow     };      _db.AuditLogs.Add(entry);     await _db.SaveChangesAsync(); }</pre>      |

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|                           | <p><b>ReservationsController.cs – Anti-forgery &amp; Authorize on actions (ReservationsController.cs 158-173)</b></p> <p>[ValidateAntiForgeryToken] on every POST; [Authorize(Roles=...)] restricts Confirm to Admin/Staff.</p> <pre>[Authorize(Roles = \$"{RoleNames.Admin},{RoleNames.Staff}")]<br/>[HttpPost, ValidateAntiForgeryToken]<br/>public async Task&lt;ActionResult&gt; Confirm(int id)<br/>{<br/>    try<br/>    {<br/>        await _service.ConfirmAsync(id, _current.UserId ?? "system");<br/>        TempData["Success"] = \$"Reservation #{id} confirmed.";<br/>    }<br/>    catch (InvalidOperationException ex) { TempData["Error"] = ex.Message; }<br/>    return RedirectToAction(nameof(Details), new { id });<br/>}</pre>                                                                                                                                                                                                                                                                                                                                                                                  |
|                           | <p><b>Program.cs – Global exception filter (Program.cs 124-127)</b></p> <p>All unhandled exceptions funnel through a single filter that logs (via IAuditLogger) and returns a generic error view – no stack traces leak to the client.</p> <pre>builder.Services.AddControllersWithViews(o =&gt;<br/>{<br/>    o.Filters.Add&lt;GlobalExceptionHandler&gt;();<br/>});</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                           | <p><b>Common/Filters/GlobalExceptionHandler.cs – Filter implementation (GlobalExceptionHandler.cs 1-22)</b></p> <pre>public class GlobalExceptionHandler : IExceptionHandler<br/>{<br/>    private readonly ILogger&lt;GlobalExceptionHandler&gt; _log;<br/>    private readonly IAuditLogger _audit;<br/>    private readonly ICurrentUser _current;<br/><br/>    public GlobalExceptionHandler(ILogger&lt;GlobalExceptionHandler&gt; log,<br/>        IAuditLogger audit,<br/>        ICurrentUser current)<br/>    {<br/>        _log = log; _audit = audit; _current = current;<br/>    }<br/><br/>    public void OnException(ExceptionContext ctx)<br/>    {<br/>        var userId = _current.UserId ?? "anonymous";<br/>        _log.LogError(ctx.Exception, "Unhandled error for user {UserId}", userId);<br/>        _audit.LogAsync(userId, "UnhandledException", "Application", "N/A",<br/>            newValues: new { Message = ctx.Exception.Message }).GetAwaiter().GetResult();<br/><br/>        ctx.Result = new ViewResult { ViewName = "Error" };<br/>        ctx.ExceptionHandled = true;<br/>    }<br/>}</pre> |
| <b>Date Submitted:</b>    | <b>September 29, 2026</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Teacher's Feedback</b> | <hr/> <hr/> <hr/> <hr/> <hr/> <hr/> <hr/> <hr/><br><div> <div>Student's Signature (after feedbacking)</div> <div>Teacher's Signature</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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